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Computer engineering (Sandip University)



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WINTER – 2019 EXAMINATION

Subject Name: Operating System

Model Answer

Subject Code: 22516

Important Instructions to examiners:

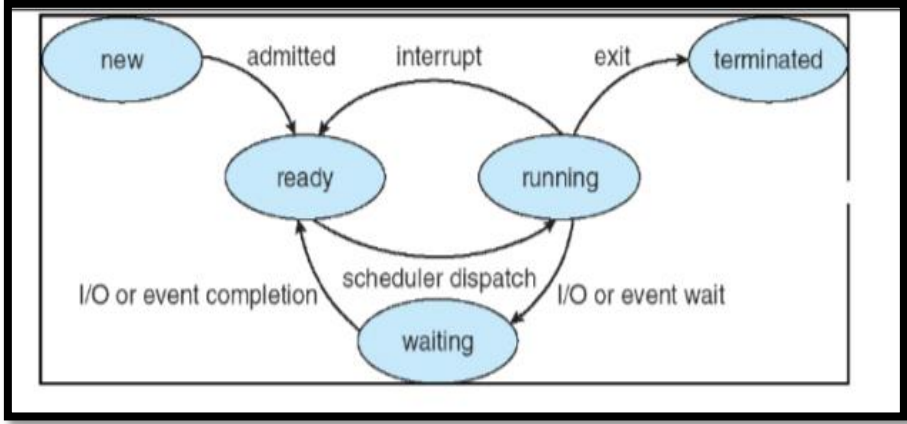
- 1) The answers should be examined by key words and not as word-to-word as given in the model answer scheme.
- 2) The model answer and the answer written by candidate may vary but the examiner may try to assess the understanding level of the candidate.
- 3) The language errors such as grammatical, spelling errors should not be given more Importance (Not applicable for subject English and Communication Skills).
- 4) While assessing figures, examiner may give credit for principal components indicated in the figure. The figures drawn by candidate and model answer may vary. The examiner may give credit for any equivalent figure drawn.
- 5) Credits may be given step wise for numerical problems. In some cases, the assumed constant values may vary and there may be some difference in the candidate's answers and model answer.
- 6) In case of some questions credit may be given by judgement on part of examiner of relevant answer based on candidate's understanding.
- 7) For programming language papers, credit may be given to any other program based on equivalent concept.

Q. No .	Sub Q. N.	Answer	Marking Scheme
1.		Attempt any Five of the following:	10 M
	a	Define real time operating system. List its any four applications of it.	2 M
	Ans	Real time Operating System: A real time system has well defined fixed time constraints. Processing should be done within the defined constraints -Hard and Soft real time system. OR The real-time operating system used for a real-time application means for those applications where data processing should be done in the fixed and small quantum of time. Types of real time operating system 1. Hard real-time 2. Soft real-time Applications: 1. Flight Control System 2. Simulations 3. Industrial control 4. Military applications	1 Mark :- Definition; 1 Mark :- for any 4 correct applications



	b	Explain any 4 services provided by OS.	2 M
	Ans	1. User Interface: All operating systems have a user interface that allows users to communicate with the system. Three types of user interfaces are available: a. Command line interface (CLI) b. Batch interface c. Graphical user interface (GUI) 2. Program execution: The operating system provides an environment where the user can conveniently run programs. It also performs other important tasks like allocation and deallocation of memory, CPU scheduling etc. It also provides service to end process execution either normally or abnormally by indicating error. 3. I/O operations: When a program is running, it may require input/output resources such as a file or devices such as printer. So the operating system provides a service to do I/O. 4. File system manipulation: Programs may need to read and write data from and to the files and directories. Operating system manages the secondary storage. Operating system makes it easier for user programs to accomplish their task such as opening a file, saving a file and deleting a file from the storage disk. 5. Communication: In the system, one process may need to exchange information with another process. Communication can be implemented via shared memory or through message passing, in which packets of information are moved between processes by the operating system. 6. Error detection: Operating systems detects CPU and memory hardware such as a memory error or power failure, a connection failure on a network or lack of paper in the printer etc. 7. Resource allocation: Operating system manages resource allocation to the processes. These resources are CPU, main memory, file storage and I/O devices. 8. Accounting: Operating system keeps track of usages of various computer resources allocated to users. 9. Protection & security: When several separate processes execute concurrently, one process should not interfere with the other processes or operating system itself. Protection provides controlled access to system resources. Security is provided by user authentication such as password for accessing information.	1 marks for explaining any 4 services



c	Draw process state diagram.	2 M
Ans	 process state diagram	2 Marks:- for correct well labelled diagram (1 mark:- specifying correct states in the diagram)
d	Explain any four scheduling criteria.	2 M
Ans	1. CPU utilization: In multiprogramming the main objective is to keep CPU as busy as possible. CPU utilization can range from 0 to 100 percent. 2.Throughput: It is the number of processes that are completed per unit time. It is a measure of work done in the system. When CPU is busy in executing processes, then work is being done in the system. Throughput depends on the execution time required for any process. For long processes, throughput can be one process per unit time whereas for short processes it may be 10 processes per unit time. 3. Turnaround time: The time interval from the time of submission of a process to the time of completion of that process is called as turnaround time. It is the sum of time period spent waiting to get into the memory, waiting in the ready queue, executing with the CPU, and doing I/O operations. 4.Waiting time: It is the sum of time periods spent in the ready queue by a process. When a process is selected from job pool, it is loaded into the main memory (ready queue). A process waits in ready queue till CPU is allocated to it. Once the CPU is allocated to the process, it starts its execution and if required request for resources. When the resources are not available that process goes into waiting state and when I/O request completes, it goes back to ready queue. In ready queue again it waits for CPU allocation. 5.Response time: The time period from the submission of a request until the first response is produced is called as response time. It is the time when system responds to the process request not the completion of	Any four scheduling criteria: 1/2 mark each



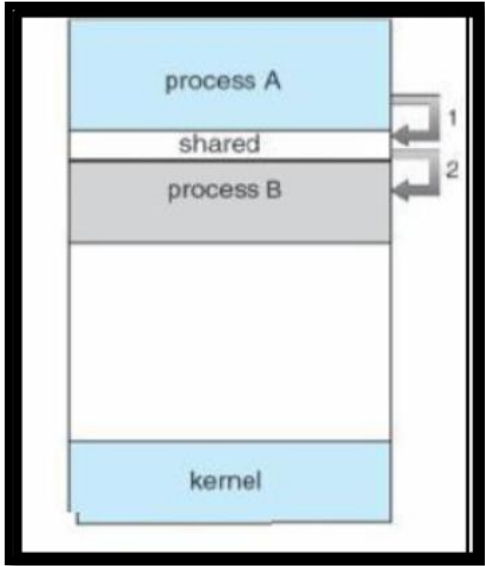
		a process. In the system, a process can Produce some output fairly early and can continue computing new results while previous results are being output to the user.	
	e	Define virtual memory	2 M
	Ans	Virtual memory is a memory management capability of an operating system (OS) that uses hardware and software to allow a computer to compensate for physical memory shortages by temporarily transferring data from random access memory (RAM) to disk storage. OR Virtual memory is the separation of user logical memory from physical memory. This separation allows an extremely large virtual memory to be provided for programmers when only a smaller physical memory is available. Virtual memory makes the task of programming much easier, because the programmer no longer needs to worry about the amount of physical memory available, or about what code can be placed in overlays, but can concentrate instead on the problem to be programmed.	2 marks for any relevant definition
	f	Write syntax for following commands: i)Sleep ii)Kill	2 M
	Ans	i)sleep Syntax: sleep NUMBER[SUFFIX]... sleep OPTION ii) kill Syntax: kill pid	1 mark each for correct syntax
	g	Describe any four file attributes	2 M
	Ans	File attributes: • Name: The symbolic file name is the only information kept in human readable form. • Identifier: File system gives a unique tag or number that identifies file within file system and which is used to refer files internally. • Type: This information is needed for those systems that support different types. • Location: This information is a pointer to a device and to the location of the file on that device. • Size: The current size of the file (in bytes, words or blocks) and possibly the maximum allowed size are included in this attribute. • Protection: Access control information determines that who can do reading, writing, executing and so on. • Time, Date and User Identification: This information may be kept for creation, Last modification and last use. These data can be useful for protection, security and usage monitoring.	Any four attributes: ½ mark each



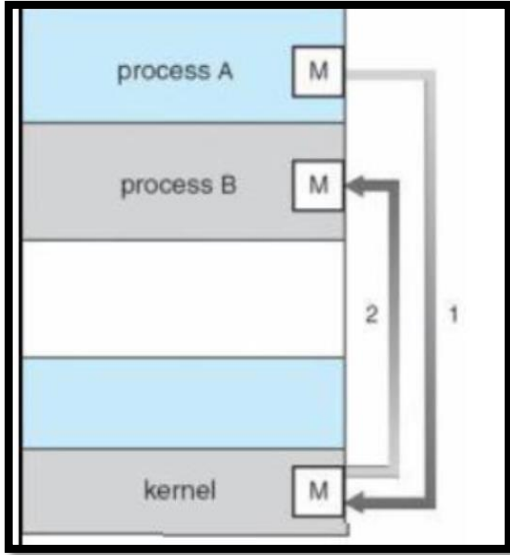
2.		Attempt any Three of the following:	12M
	a	Enlist types of operating system. Explain multiprogramming OS in detail.	4M
	Ans	Types of operating system 1. Batch Systems 2. Multiprogramming 3. Multitasking 4. Time-Sharing Systems 5. Desktop Systems 6. Distributed system 7. Clustered system 8. Real Time system: Multiprogramming: • In multiprogramming, more than one program lies in the memory. • The scheduler selects the jobs to be placed in ready queue from a number of programs. • The ready queue is placed in memory and the existence of more than one program in main memory is known as multiprogramming. • Since there is only one processor, there multiple programs cannot be executed at a time. • Instead the operating system executes part of one program, then the part of another and so on. • Example of multiprogramming: user can open word, excel, access and other applications in a system. Fig- Multiprogramming with three programs	1 Mark:- Listing; 1 Mark:- Diagram 2 Marks:- Explanation
	b	List components of OS. Explain process management in detail.	4M
	Ans	List of System Components: 1. Process management 2. Main memory management 3. File management 4. I/O system management 5. Secondary storage management	1 Mark:- Listing; 3 Marks:- Explanation



		Process Management: The operating system manages many kinds of activities ranging from user programs to system programs like printer spooler, name servers, file server etc. Each of these activities is encapsulated in a process. • A process includes the complete execution context (code, data, PC, registers, OS resources in use etc.). • The basic unit of software that the operating system deals with in scheduling the work done by the processor is either a process or a thread, depending on the operating system. • It's tempting to think of a process as an application, but that gives an incomplete picture of how processes relate to the operating system and hardware. • The application you see (word processor or spreadsheet or game) is, indeed, a process, but that application may cause several other processes to begin, for tasks like communications with other devices or other computers. • There are also numerous processes that run without giving you direct evidence that they ever exist. A process, then, is software that performs some action and can be controlled by a user, by other applications or by the operating system. • It is processes, rather than applications, that the operating system controls and schedules for execution by the CPU. In a single-tasking system, the schedule is straightforward. • The operating system allows the application to begin running, suspending the execution only long enough to deal with interrupts and user input. • The five major activities of an operating system in regard to process management are 1. Creation and deletion of user and system processes. 2. Suspension and resumption of processes. 3. A mechanism for process synchronization. 4. A mechanism for process communication. 5. A mechanism for deadlock handling.	
	c	With neat diagram explain inter process communication model.	4M
	Ans	Inter-process communication: Cooperating processes require an Inter-process communication (IPC) mechanism that will allow them to exchange data and information. There are two models of IPC 1. Shared memory	Define inter process communication -1 mark; diagram of model - 1 mark; explanation

		 • In this, all processes who want to communicate with other processes can access a region of the memory residing in an address space of a process creating a shared memory segment. • All the processes using the shared memory segment should attach to the address space of the shared memory. All the processes can exchange information by reading and/or writing data in shared memory segment. • The form of data and location are determined by these processes who want to communicate with each other. • These processes are not under the control of the operating system. • The processes are also responsible for ensuring that they are not writing to the same location simultaneously. • After establishing shared memory segment, all accesses to the shared memory segment are treated as routine memory access and without assistance of kernel. 2. Message Passing • In this model, communication takes place by exchanging messages between cooperating processes. • It allows processes to communicate and synchronize their action without sharing the same address space. • It is particularly useful in a distributed environment when communication process may reside on a different computer connected by a network. • Communication requires sending and receiving messages through the kernel.	- 2 marks
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		 The processes that want to communicate with each other must have a communication link between them. Between each pair of processes exactly one communication link.	
	d	Describe I/O Burst and CPU Burst cycle with neat diagram.	4M
	Ans	CPU burst cycle: It is a time period when process is busy with CPU. I/O burst cycle: It is a time period when process is busy in working with I/O resources. - A process execution consists of a cycle of CPU execution and I/O wait. - A process starts its execution when CPU is assigned to it, so process execution begins with a CPU burst cycle. - This is followed by an I/O burst cycle when a process is busy doing I/O operations. - A process switch frequently from CPU burst cycle to I/O burst cycle and vice versa. - The complete execution of a process starts with CPU burst cycle, followed by I/O burst cycle, then followed by another CPU burst cycle, then followed by another I/O burst cycle and so on. - The final CPU burst cycle ends with a system request to terminate execution.	Explanation: 2 marks, Diagram: 2 marks

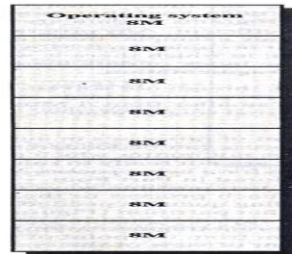
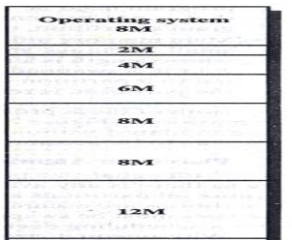


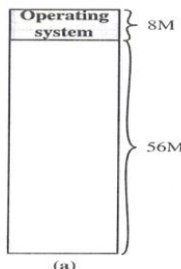
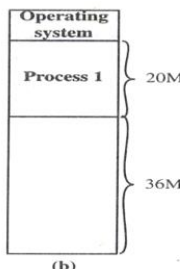
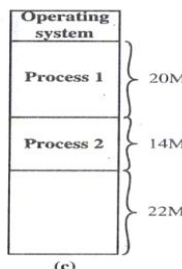
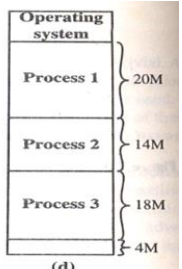
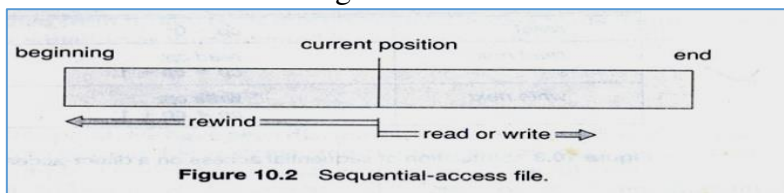
3.		Attempt any Three of the following:	12M
	a	Explain 'PS' command with any four options.	4M
	Ans	ps command: It is used to display the characteristics of a process. This command when execute without options, it lists the processes associated with a user at a particular terminal. Syntax: \$ ps [options] Example: \$ ps output: <pre> PID TTY TIME CMD 12330 pts/0 00:00:00 bash 21621 pts/0 00:00:00 ps </pre> Each line in the output shows PID, the terminal with which the process is associated, the cumulative processor time that has been consumed since the process has been started and the process name. Options: -f : It is used to display full listing of attributes of a process. It includes UID (user ID),PPID(Parent ID),C(amount of CPU time consumed by the process) and STIME(chronological time that has elapsed since the process started). Example: \$ ps -f <pre> UID PID PPID C STIME TTY TIME CMD root 1 0 0 19:58 ? 00:00:01 /sbin/ini root 2 0 0 19:58 ? 00:00:00 [kthreadd root 3 2 0 19:58 ? 00:00:00 [ksoftirq </pre> -u: Shows the activities of any specified user at any time. Example: \$ ps -u abc	four options-1M each



	<pre>PID TTY TIME CMD 1053 ? 00:00:00 systemd 1062 ? 00:00:00 (sd-pam) 1074 tty1 00:00:00 zsh</pre> -a: It shows the processes of all users. Example: \$ ps -a <pre>PID TTY TIME CMD 27011 pts/0 00:00:00 man 27016 pts/0 00:00:00 less 27499 pts/1 00:00:00 ps</pre> -e: It displays processes including user and system processes. example: \$ ps -e <pre>PID TTY TIME CMD 1 ? 00:00:05 systemd 2 ? 00:00:00 kthreadd 3 ? 00:00:00 ksoftirqd/0 5 ? 00:00:00 kworker/0:0H 7 ? 00:00:01 rcu_sched 8 ? 00:00:00 rcu_bh</pre>	
b	Explain deadlock? What are necessary conditions for deadlock?	4M
Ans	In multiprogramming environment, several processes may compete for a finite number of resources. A process requests resources and if the resources are not available then the process enters into the waiting state. Sometimes a waiting process is never again able to change its status because the resources requested by it are held by other waiting processes. This situation is called as deadlock. When a process request for resources held by another waiting process which in turn is waiting for resources held by another waiting process and not a single process can execute its task, then deadlock occurs in the system. Example: Consider a system with three disk drives and three processes. When each process request one disk drive, system allocates one disk drive to each process. Now there is no more drive available in the system. If all three processes request for one more disk drive, then all three processes will go into the waiting state and system will go in deadlock state. Because any one process from the three can execute only when one of them will release the disk drive allocated to it. Necessary Conditions: 1. Mutual exclusion: At least one resource must be held in a non-sharable mode; that is, only one process at a time can use the resource. 2. Hold and Wait: A process must be holding at least one resource and waiting to acquire additional resources that are currently being held by	Deadlock description- 2M, necessary conditions - 1/2 M each



	other processes.	
	3. No pre-emption: Resources cannot be pre-empted i.e a resource can be released only voluntarily by the process holding it. 4. Circular wait: A set $\{P_0, P_1, \dots, P_n\}$ of waiting processes must exist such that P_0 is waiting for a resource held by P_1, P_1 is waiting for a resource held by P_2, ..., P_{n-1} is waiting for a resource held by P_n and P_n is waiting for a resource held by P_0. Each process is waiting for the resources held by other waiting processes in circular form.	
c	Explain partitioning and its types.	4M
Ans	An important operation of memory management is to bring programs into main memory for execution by the processor. Partitioning is a technique that divides a memory into multiple partitions. These partitions can be of different size or same size. Types of partitioning - Fixed partitioning i.e. static partitioning - Variable partitioning i.e. dynamic partitioning Fixed Partitioning: Main memory is divided into multiple partitions of fixed size at the time of system generation. A process may be loaded into a partition of equal size or greater size. Partitions can be of equal size or unequal size. Equal size partitioning: Main memory is divided into equal size partitions. Any process with less or equal size can be loaded in any available partition.  OR Unequal size partitioning: Main memory is divided into multiple partitions of unequal size. Each process can be loaded into the smallest partition within which the process will fit.  Variable partitioning: When a process enters in main memory, it is allocated exact size that is required by that process. So in this method, partitions can vary in size depending on memory space required by a process entering in main memory. Operating system maintains a table indicating which parts of memory are available and which are occupied. When new process arrives and it needs space, system searches for	Explanation of fixed partitioning -2M, Variable partitioning-2M

	available memory space in main memory. If it is available, then memory is allocated to the process by creating a partition in memory. For example: Consider following table with process and memory space. <table><tr><th>Process</th><th>Memory space</th></tr><tr><td>P1</td><td>20 M</td></tr><tr><td>P2</td><td>14 M</td></tr><tr><td>P3</td><td>18 M</td></tr></table>  (a)  (b)  (c)  (d)	Process	Memory space	P1	20 M	P2	14 M	P3	18 M	
Process	Memory space									
P1	20 M									
P2	14 M									
P3	18 M									
d	Describe sequential and direct access method.	4M								
Ans	Sequential access: Information from the file is processed in order i.e. one record after another. It is commonly used access mode. For example, editors and compilers access files in sequence. A read operation read information from the file in a sequence i.e. read next reads the next portion of the file and automatically advances a file pointer. A write operation writes information into the file in a sequence i.e. write next appends to the end of the file and advances to the end of the newly written material. Such a file can be reset to the beginning. In some operating systems, a program may be able to skip forward or backward n records for some integer n.  Figure 10.2 Sequential-access file. As shown in above diagram, a file can be rewind (moved in backward direction) from the current position to start with beginning of the file or it can be read or write in forward direction. Direct access: It is also called as relative access. A file is made up of fixed length logical records that allow programs to read and write records rapidly in no particular order. Direct access method is based on disk model of a file which allows random access to any file block. For direct access a file is viewed as a numbered sequence of blocks or records. So we can directly read block 14, then block 53 and so on. This method is used for immediate access to large amount of information. Database can be accessed with direct access method. For example, when	description of sequential access-2M, Direct access-2M								



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		a query concerning a particular subject arrives, we compute which block contains the answer and then read that block directly to provide the desired information. Read n operation is used to read the nth block from the file whereas write n is used to write in that block. The block numbers provided by the user to the operating system is a relative block number. A relative block number is an index relative to the beginning of the file. The first relative block of file is 0; the next is 1 and so on. Actual absolute disk address of the block is different from the relative address. The use of relative block numbers allow the operating system to decide where the file should be placed and helps to prevent the user from accessing portions of the file system that may not be part of his file.	
4		Attempt any Three of the following:	12M
	a	Write Unix command for following: i) create a folder OSY ii) create a file FIRST in OSY folder iii) List/display all files and directories. iv) Write command to clear the screen	4M
	Ans	i) create a folder OSY: \$mkdir OSY ii) create a file FIRST in OSY folder: \$cd OSY \$cat>FIRST or \$ touch FIRST iii) List/display all files and directories: \$ls iv) to clear screen: \$clear	Each correct command-1M
	b	What is purpose of system call? State two system calls with their functions.	4M
	Ans	System call provides an interface between a running program and operating system. It allows user to access services provided by operating system. This system calls are procedures written using C, C++ and assembly language instructions. Each operating system has its own name for each system call. Each system call is associated with a number that identifies itself. System calls: Process Control: Program in execution is a process. A process to be executed must be loaded in main memory. while executing it may need to wait, terminate or create & terminate child processes. • end, abort • load, execute • create process, terminate process	purpose of system call-2M, Two system calls-1M each



	• get process attributes, set process attributes • wait for time • wait event, signal event • allocate and free memory File Management: System allows us to create and delete files. For create and delete operation system call requires the name of the file and other attributes of the file. File attributes include file type, file size, protection codes, accounting information and so on. Systems access these attributes for performing operations on file and directories. Once the file is created, we can open it and use it. System also allows performing reading, writing or repositioning operations on file. • create file, delete file • open, close • read, write, reposition • get file attributes, set device attributes • logically attach or detach devices 3. Device Management: When a process is in running state, it requires several resources to execute. These resources include main memory, disk drives, files and so on. If the resource is available, it is assigned to the process. Once the resource is allocated to the process, process can read, write and reposition the device. • request device, release device • read, write, reposition • get device attributes, set device attributes • logically attach or detach devices 4. Information Maintenance: Transferring information between the user program and the operating system requires system call. System information includes displaying current date and time, the number of current user, the version number of the operating system, the amount of free memory or disk space and so on. Operating system keeps information about all its processes that can be accessed with system calls such as get process attributes and set process attributes. • get time or date, set time or date • get system data, set system data • get process, file, or devices attributes • set process, file, or devices attributes 5. Communication: Processes in the system, communicate with each other. Communication is done by using two models: message passing and shared memory. For transferring messages, sender process connects itself to receiving process by specifying receiving process	
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		name or identity. Once the communication is over system close the connection between communicating processes. • create, delete communication connection • send, receive messages • transfer status information • attach or detach remote devices.	
	c	State and describe types of scheduler.	4M
	Ans	There are three types of scheduler: • Long term scheduler • Short term scheduler • Medium term scheduler 1. Long term scheduler: It selects programs from job pool and loads them into the main memory. It controls the degree of multiprogramming. The degree of multiprogramming is the number of processes loaded (existing) into the main memory. System contains I/O bound processes and CPU bound processes. An I/O bound process spends more time for doing I/O operations whereas CPU bound process spends more time in doing computations with the CPU. So It is the responsibility of long term scheduler to balance the system by loading some I/O bound and some CPU bound processed into the main memory. Long term scheduler executes only when a process leaves the system, so it executes less frequently. When long term scheduler selects a process from job pool, the state of process changes from new to ready state. 2. Short term scheduler: It is also known as CPU scheduler. This scheduler selects processes that are ready for execution from the ready queue and allocates the CPU to the selected process. Frequency of execution of short term scheduler is more than other schedulers. When short term scheduler selects a process, the state of process changes from ready to running state. 3. Medium term scheduler: When a process is in running state, due to some interrupt it is blocked. System swaps out blocked process and store it into a blocked and swapped out process queue. When space is available in the main memory, the operating system looks at the list of swapped out but ready processes. The medium term scheduler selects one process from that list and loads it into the ready queue. The job of medium term scheduler is to select a process from swapped out process queue and to load it into the main memory. This scheduler works in close communication with long term scheduler for loading process into the main memory.	list-1M, description of each-1
	d	Explain Round Robin algorithm with suitable example.	4M



	Ans It is preemptive scheduling algorithm. A small unit of time known as a time quantum or time slice is used for pre-emption of a currently running process. Ready queue is implemented as a circular queue. CPU is assigned to the entire processes one by one, on first come first serve basis, for a specific time period. Every process executes for specified time period and CPU is given to the next process when time quantum expires. A new process is added at the tail of the ready queue when it enters the system. CPU scheduler selects first process from head of the ready queue and executes it for a specified time quantum. Once the time quantum expires, dispatcher is invoked to pre-empt current running process and CPU is given to the next process placed at the head of the ready queue. The running process may have a CPU burst time less or greater than time quantum. If burst time of running process is less than the time quantum then, the process itself releases the CPU. The scheduler then selects next process from ready queue and executes it. If burst time of running process is longer than time quantum then, context switch occurs and the process is place at the tail of ready queue for remaining burst time execution. Example: <table> <tr> <td><u>Process</u></td> <td><u>Burst Time</u></td> </tr> <tr> <td>P₁</td> <td>24</td> </tr> <tr> <td>P₂</td> <td>3</td> </tr> <tr> <td>P₃</td> <td>3</td> </tr> </table> Time quantum: 4 ms The resulting RR schedule is as follows: <table> <tr> <td>P₁</td> <td>P₂</td> <td>P₃</td> <td>P₁</td> <td>P₁</td> <td>P₁</td> <td>P₁</td> <td>P₁</td> </tr> <tr> <td>0</td> <td>4</td> <td>7</td> <td>10</td> <td>14</td> <td>18</td> <td>22</td> <td>26</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>30</td> </tr> </table> CPU is allocated to process P₁ for 4 ms. Since it requires another 20 milliseconds, it is preempted after the first time quantum and the CPU is given to the next process in the queue, process P₂. Process P₂ does not need 4 milliseconds, so it quits before its time quantum expires. The CPU is then given to the next process, process P₃. Once each process has received 1 time quantum, the CPU returns to process P₁ for an additional time quantum.	<u>Process</u>	<u>Burst Time</u>	P ₁	24	P ₂	3	P ₃	3	P ₁	P ₂	P ₃	P ₁	P ₁	P ₁	P ₁	P ₁	0	4	7	10	14	18	22	26								30	explanation of round robin -2M, example-2M
<u>Process</u>	<u>Burst Time</u>																																	
P ₁	24																																	
P ₂	3																																	
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P ₁	P ₂	P ₃	P ₁	P ₁	P ₁	P ₁	P ₁																											
0	4	7	10	14	18	22	26																											
							30																											
	e Explain PCB with diagram.	4M																																
	Ans Each process is represented as a process control block (PCB) in the operating system. It contains information associated with specific process. Process State: It indicates current state of a process. Process state can be new, ready, running, waiting and terminated. Process number: Each process is associated with a unique number	explanation-2M, diagram-2M																																

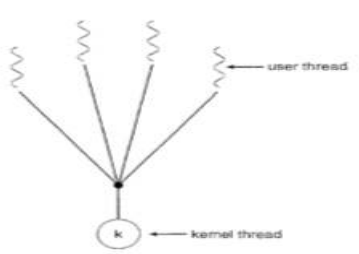


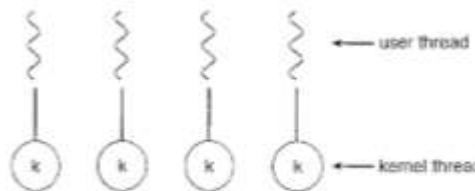
		which is known process identification number. Program Counter: It indicates the address of the next instruction to be executed for the process. CPU Registers: The registers vary in number and type depending on the computer architecture. Register includes accumulators, index registers, stack pointers and general purpose registers plus any condition code information. Memory Management Information: It includes information such as value of base and limit registers, page tables, segment tables, depending on the memory system used by OS. Accounting Information: This information includes the amount of CPU used, time limits, account holders, job or process number and so on. It also includes information about listed I/O devices allocated to the process such as list of open files. Each PCB gives information about a particular process for which it is designed. process state process number program counter registers memory limits list of open files ...	
5		Attempt any Two of the following:	12M
	a	Enlist the operating system tools. Explain any two in detail.	6M
	Ans	Following are the operating tools: • User Management • Security policy • Device Management • Performance Monitor • Task Scheduler A) User management: • User management includes everything from creating a user to deleting a user on your system. User management can be done in three ways on a Linux system. • Command line tools include commands like useradd, userdel, usermod, passwd, etc. These are mostly used by the server administrators. Useradd: With useradd commands you can add a user.	For List=2 Marks and Explanation any two for 4 Marks



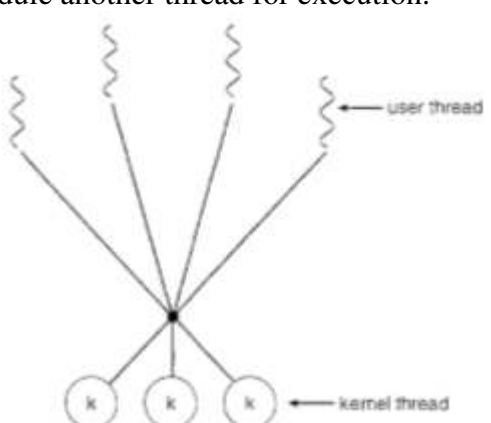
	Syntax: <code>useradd -m -d /home/<userName> -c "<userName>" <userName></code> Example: <code>useradd -m -d /home/xyz -c "xyz" xyz</code> File <code>/etc/default/useradd</code> contains some user default options. The command <code>useradd -D</code> can be used to display this file. Syntax: <code>useradd -D</code> Userdel: To delete a user account <code>userdel</code> command is used. Syntax: <code>userdel -r <userName></code> Usermod: The command <code>usermod</code> is used to modify the properties of an existing user. Syntax: <code>usermod -c '<newName>' <oldName></code> Example: <code>usermod -c 'vppoly' john</code> Using passwd command Passwd: A user can set the password with the command passwd. Old password has to be typed twice before entering the new one. Syntax: <code>passwd <userName></code> Example: <code>passwd vppoly</code> B) Device Management: Device management is the process of managing the implementation, operation and maintenance of a physical and/or virtual device. All Linux device files are located in the <code>/dev</code> directory, which is an integral part of the root (<code>/</code>) filesystem because these device files must be available to the operating system during the boot process. Example: <code>ls -l /dev</code> Above example gives the list of device file from kernel. Udev supplies a dynamic device directory containing only the nodes for devices which are connected to the system. It creates or removes the device node files in the <code>/dev</code> directory. C) Performance Monitor: It is very tough job for every system or network administrator to monitor and debug Linux System Performance problems every day. The commands discussed below are some of the most fundamental commands when it comes to system analysis and debugging Linux server issues such as: 1) vmstat: Virtual memory statistics The <code>vmstat</code> command reports information about processes, memory, paging, block IO, traps, and cpu activity. \$ <code>vmstat 3</code> 2)top: Process activity monitoring command	
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		top command display Linux processes. It provides a dynamic real-time view of a running system i.e. actual process activity. By default, it displays the most CPU-intensive tasks running on the server and updates the list every five seconds. \$ top 3) free: Show Linux server memory usage free command shows the total amount of free and used physical and swap memory in the system, as well as the buffers used by the kernel. # free 4) iostat: Monitor Linux average CPU load and disk activity iostat command report Central Processing Unit (CPU) statistics and input/output statistics for devices, partitions and network filesystems (NFS). # iostat 5) netstat Linux network and statistics monitoring tool netstat command displays network connections, routing tables, interface statistics, masquerade connections, and multicast memberships. # netstat -tulpn	
	b	Explain multithreading model in detail.	6M
	Ans	Many systems provide support for both user and kernel threads, resulting in different multithreading models. Following are three multithreading model: Many-to-One Model • The many-to-one model maps many user-level threads to one kernel thread. • Thread management is done by the thread library in user space, so it is efficient; but the entire process will block if a thread makes a blocking system call. • Also, because only one thread can access the kernel at a time, multiple threads are unable to run in parallel on multiprocessors. • Example: Green threads- a thread library available for Solaris  The diagram illustrates the Many-to-One Multithreading Model. It shows four wavy lines representing user threads at the top, each with an arrow pointing downwards to a single circle at the bottom labeled 'k'. An arrow points from the text 'user thread' to one of the wavy lines, and another arrow points from the text 'kernel thread' to the circle 'k'.	Each model=2M

	Advantages: • More concurrency because of multiple threads can run in parallel on multiple CPUs. • Less complication in the processing. Disadvantages: • Thread creation involves light-weight process creation. • Kernel thread is an overhead. • Limiting the number of total threads. One-to-One Model • The one-to-one model maps each user thread to a kernel thread. • It provides more concurrency than the many-to-one model by allowing another thread to run when a thread makes a blocking system call; it also allows multiple threads to run in parallel on multiprocessors. • The only drawback to this model is that creating a user thread requires creating the corresponding kernel thread. • Because the overhead of creating kernel threads can burden the performance of an application, most implementations of this model restrict the number of threads supported by the system. • Linux, along with the family of Windows operating systems, implement the one-to-one model.  Advantages: • Mainly used in language system, portable libraries. • One kernel thread controls multiple user thread. Disadvantages: • Parallelism is not supported by this model. • One block can blocks all user threads. Many-to-Many Model • The many-to-many model multiplexes many user-level threads to a smaller or equal number of kernel threads. • The number of kernel threads may be specific to either a particular application or a particular machine (an application may be allocated more kernel threads on a multiprocessor than on a uniprocessor).	
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		- The one-to-one model allows for greater concurrency, but the developer has to be careful not to create too many threads within an application (and in some instances may be limited in the number of threads she can create). - The many-to-many model suffers from neither of these shortcomings: developers can create as many user threads as necessary, and the corresponding kernel threads can run in parallel on a multiprocessor. - Also, when a thread performs a blocking system call, the kernel can schedule another thread for execution.  Advantages: - Many threads can be created as per user's requirement. - Multiple kernel or equal to user threads can be created. Disadvantages: - True concurrency cannot be achieved. - Multiple threads of kernel is an overhead for operating system	
	c	Explain LRU page replacement algorithm for following reference string. 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 Calculate the page fault.	6M
	Ans	LRU: - The Least Recently Used (LRU) page replacement policy replaces the page that has not been used for the longest period of time. - LRU replacement associates with each page the time of that page's last use. - When a page must be replaced, LRU chooses the page that has not been used for the longest period of time. - The LRU policy is often used as a page-replacement algorithm and is considered to be good. - An LRU page-replacement algorithm may require substantial hardware assistance.	LRU explanation = 2M Calculation = 4 M



		Counters: • In the simplest case, we associate with each page-table entry a time-of-use field and add to the CPU a logical clock or counter. • The clock is incremented for every memory reference. • Whenever a reference to a page is made, the contents of the clock register are copied to the time-of-use field in the page-table entry for that page. • In this way, we always have the "time" of the last reference to each page. We replace the page with the smallest time value. Stack: • Another approach to implementing LRU replacement is to keep a stack of page numbers. • Whenever a page is referenced, it is removed from the stack and put on the top. • In this way, the most recently used page is always at the top of the stack and the least recently used page is always at the bottom. Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 (Frame size have not mentioned in question so assume frame size as 3 or 4) LRU: Assume frame size=3 <table><tr><td>7</td><td>0</td><td>1</td><td>2</td><td>0</td><td>3</td><td>0</td><td>4</td><td>2</td><td>3</td><td>0</td><td>3</td><td>2</td><td>1</td><td>2</td><td>0</td><td>1</td><td>7</td><td>0</td><td>1</td></tr><tr><td>7</td><td>7</td><td>7</td><td>2</td><td></td><td>2</td><td></td><td>4</td><td>4</td><td>4</td><td>0</td><td></td><td></td><td>1</td><td></td><td>1</td><td>*</td><td>1</td><td></td><td>*</td></tr><tr><td></td><td>0</td><td>0</td><td>0</td><td>*</td><td>0</td><td>*</td><td>0</td><td>0</td><td>3</td><td>3</td><td>*</td><td></td><td>3</td><td></td><td>0</td><td></td><td>0</td><td>*</td><td></td></tr><tr><td></td><td></td><td></td><td>1</td><td>1</td><td></td><td>3</td><td></td><td>3</td><td>2</td><td>2</td><td>2</td><td></td><td>*</td><td>2</td><td>*</td><td>2</td><td></td><td>7</td><td></td></tr></table> Page Fault=12 Assume frame size=4 <table><tr><td>7</td><td>0</td><td>1</td><td>2</td><td>0</td><td>3</td><td>0</td><td>4</td><td>2</td><td>3</td><td>0</td><td>3</td><td>2</td><td>1</td><td>2</td><td>0</td><td>1</td><td>7</td><td>0</td><td>1</td></tr><tr><td>7</td><td>7</td><td>7</td><td>7</td><td></td><td>3</td><td></td><td>3</td><td></td><td>*</td><td></td><td>*</td><td></td><td>3</td><td></td><td></td><td></td><td>7</td><td></td><td></td></tr><tr><td></td><td>0</td><td>0</td><td>0</td><td>*</td><td>0</td><td>*</td><td>0</td><td></td><td></td><td>*</td><td></td><td></td><td>0</td><td></td><td>*</td><td></td><td>0</td><td>*</td><td></td></tr><tr><td></td><td></td><td>1</td><td>1</td><td></td><td>1</td><td></td><td>4</td><td></td><td></td><td></td><td></td><td></td><td>1</td><td></td><td></td><td>*</td><td>1</td><td></td><td>*</td></tr><tr><td></td><td></td><td></td><td>2</td><td></td><td>2</td><td></td><td>2</td><td>*</td><td></td><td></td><td></td><td>*</td><td>2</td><td>*</td><td></td><td></td><td>2</td><td></td><td></td></tr></table> Page fault=08	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1	7	7	7	2		2		4	4	4	0			1		1	*	1		*		0	0	0	*	0	*	0	0	3	3	*		3		0		0	*					1	1		3		3	2	2	2		*	2	*	2		7		7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1	7	7	7	7		3		3		*		*		3				7				0	0	0	*	0	*	0			*			0		*		0	*				1	1		1		4						1			*	1		*				2		2		2	*				*	2	*			2			
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	a	The jobs are scheduled for execution as follows:	6M SJF=3 m FCFS=3 m (1m-gantt chart, 2m calculation)																																																																																																																																																																																				



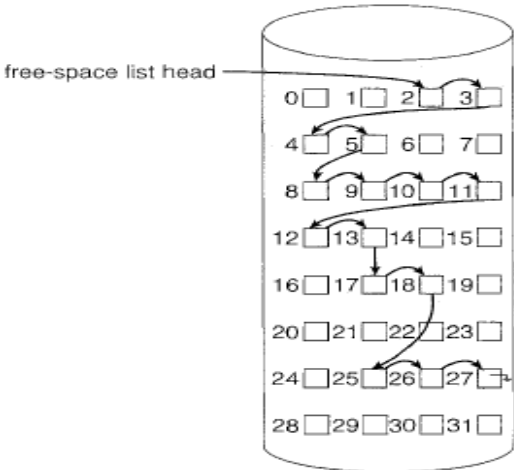
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(Autonomous)
(ISO/IEC - 27001 - 2013 Certified)

		<table><tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td></tr><tr><td>P1</td><td>0</td><td>7</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>10</td></tr><tr><td>P4</td><td>3</td><td>6</td></tr><tr><td>P5</td><td>4</td><td>8</td></tr></table> i) SJF ii) FCFS Also find average waiting time using Gantt chart.	Process	Arrival Time	Burst Time	P1	0	7	P2	1	4	P3	2	10	P4	3	6	P5	4	8	of AWT)																																																					
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Ans	SJF: <u>Non-Preemptive SJF</u> Gantt Chart: <table><tr><td>P1</td><td>P2</td><td>P4</td><td>P5</td><td>P3</td></tr><tr><td>0</td><td>7</td><td>11</td><td>17</td><td>25</td><td>35</td></tr></table> <table><tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td><td>Waiting Time</td></tr><tr><td>P1</td><td>0</td><td>7</td><td>0</td></tr><tr><td>P2</td><td>1</td><td>4</td><td>7-1=6</td></tr><tr><td>P3</td><td>2</td><td>10</td><td>25-2=23</td></tr><tr><td>P4</td><td>3</td><td>6</td><td>11-3=8</td></tr><tr><td>P5</td><td>4</td><td>8</td><td>17-4=13</td></tr></table> Average waiting Time=(0+6+23+08+13)/5 = 50/5=10 OR <u>Preemptive SJF</u> Gantt Chart: <table><tr><td>P1</td><td>P2</td><td>P1</td><td>P4</td><td>P5</td><td>P3</td></tr><tr><td>0</td><td>1</td><td>5</td><td>11</td><td>17</td><td>25</td><td>35</td></tr></table> <table><tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td><td>Waiting Time</td></tr><tr><td>P1</td><td>0</td><td>7</td><td>0+(5-1)=4</td></tr><tr><td>P2</td><td>1</td><td>4</td><td>1-1=0</td></tr><tr><td>P3</td><td>2</td><td>10</td><td>25-2=23</td></tr><tr><td>P4</td><td>3</td><td>6</td><td>11-3=8</td></tr><tr><td>P5</td><td>4</td><td>8</td><td>17-4=13</td></tr></table> Average Waiting Time=4+0+23+8+13/5=9.6	P1	P2	P4	P5	P3	0	7	11	17	25	35	Process	Arrival Time	Burst Time	Waiting Time	P1	0	7	0	P2	1	4	7-1=6	P3	2	10	25-2=23	P4	3	6	11-3=8	P5	4	8	17-4=13	P1	P2	P1	P4	P5	P3	0	1	5	11	17	25	35	Process	Arrival Time	Burst Time	Waiting Time	P1	0	7	0+(5-1)=4	P2	1	4	1-1=0	P3	2	10	25-2=23	P4	3	6	11-3=8	P5	4	8	17-4=13	Note:
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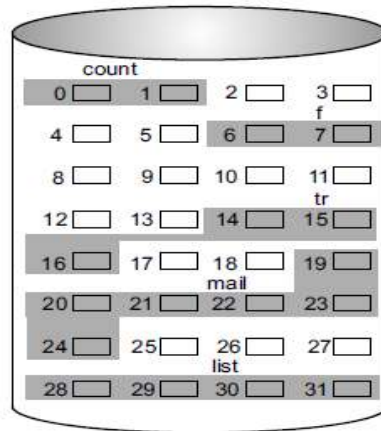
		ii) FCFS Gantt chart: <table><tr><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td></tr><tr><td>0</td><td>7</td><td>11</td><td>21</td><td>27</td><td>35</td></tr></table> <table><tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td><td>Waiting Time</td></tr><tr><td>P1</td><td>0</td><td>7</td><td>0-0=0</td></tr><tr><td>P2</td><td>1</td><td>4</td><td>7-1=6</td></tr><tr><td>P3</td><td>2</td><td>10</td><td>11-2=9</td></tr><tr><td>P4</td><td>3</td><td>6</td><td>21-3=18</td></tr><tr><td>P5</td><td>4</td><td>8</td><td>27-4=23</td></tr></table> Average waiting Time=0+6+9+18+23/5 =56/5=11.2	P1	P2	P3	P4	P5	0	7	11	21	27	35	Process	Arrival Time	Burst Time	Waiting Time	P1	0	7	0-0=0	P2	1	4	7-1=6	P3	2	10	11-2=9	P4	3	6	21-3=18	P5	4	8	27-4=23	
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P4	3	6	21-3=18																																			
P5	4	8	27-4=23																																			
	b	List free space management techniques? Describe any one in detail.	6M																																			
	Ans	A file system is responsible to allocate the free blocks to the file therefore it has to keep track of all the free blocks present in the disk. There are mainly four approaches by using which, the free blocks in the disk are managed. • Bit Vector • Linked List • Grouping • Counting Bit Vector: • The free-space list is implemented as a bit map or bit vector. • Each block is represented by 1 bit. If the block is free, the bit is 1; if the block is allocated, the bit is 0. • For example, consider a disk where blocks • 2, 3, 4, 5, 8, 9, 10, 11, 12, 13 are free and the rest of the blocks are allocated. • The free-space bit map would be : 001111001111100 <table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> 0=Free block 1= Allocated block The main advantage of this approach is its relative simplicity and its efficiency in finding the first free block or n consecutive free blocks on the disk.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0	Listing=1M Explanation =3M And Diagram=2M			
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15																							
0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0																							



		Linked List • In this approach, the free disk blocks are linked together i.e. a free block contains a pointer to the next free block. • The block number of the very first disk block is stored at a separate location on disk and is also cached in memory. • In this approach, link all the disk blocks together, keeping a pointer to the first free block. • This block contains a pointer to the next free disk block, and so on. 	
	c	Enlist different file allocation methods? Explain contiguous allocation method in detail.	
	Ans	From the user's point of view, a file is an abstract data type. It can be created, opened, written, read, closed and deleted without any real concern for its implementation. The implementation of a file is a problem for the operating system. The main problem is how to allocate space to these files so that disk space is effectively utilized and files can be quickly accessed. Three major methods of allocating disk space are in wide use: • Contiguous • Linked • Indexed Contiguous Allocation • The contiguous allocation method requires each file to occupy a set of contiguous addresses on the disk. Disk addresses define a linear ordering on the disk. Contiguous allocation of a file is defined by the disk address of the first block and its length. If the file is 'n' blocks long and starts at location 'b', then it occupies blocks b, b+1, b+2, - - - - b+n-1. The directory entry for each file indicates the address of the starting block and the length of the area allocated for this file. • Contiguous allocation supports both sequential and direct	1m- listing, 2m for diagram, 3m for explanation

access.

- For direct access to block 'i' of a file, which starts at block 'b', we can immediately access block $b+i$. The difficulty with contiguous allocation is finding space for a new file.
- For direct access to block 'i' of a file, which starts at block 'b', we can immediately access block $b+i$.
- The difficulty with contiguous allocation is finding space for a new file.
- If file to be created are 'n' blocks long, we must search free space list for 'n' free contiguous blocks.



Directory		
File	Start	Length
count	0	2
tr	14	3
mail	19	6
list	28	4
f	6	2

Advantages of Contiguous File Allocation Method:

1. Supports both sequential and direct access methods.
2. Contiguous allocation is the best form of allocation for sequential files. Multiple blocks can be brought in at a time to improve I/O performance for sequential processing.
3. It is also easy to retrieve a single block from a file. For example, if a file starts at block 'n' and the ith block of the file is wanted, its location on secondary storage is simply $n + i$.
4. Reading all blocks belonging to each file is very fast.
5. Provides good performance.

Disadvantages of Contiguous File Allocation Method:

1. Suffers from external fragmentation.
2. Very difficult to find contiguous blocks of space for new files.
3. Also with pre-allocation, it is necessary to declare the size of the file at the time of creation which many a times is difficult to estimate.
4. Compaction may be required and it can be very expensive.

22516

21222

3 Hours / 70 Marks

Seat No.

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15 minutes extra for each hour

- Instructions* – (1) All Questions are *Compulsory*.
- (2) Answer each next main Question on a new page.
- (3) Illustrate your answers with neat sketches wherever necessary.
- (4) Figures to the right indicate full marks.
- (5) Assume suitable data, if necessary.
- (6) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following: 10
- a) State and describe any two advantages of multiprocessor system.
- b) State any four types of system calls.
- c) Draw a neat labelled diagram for process state.
- d) State difference between preemptive scheduling and non-preemptive scheduling.
- e) Define the term fragmentation in terms of memory.
- f) Give commands to perform following tasks:
- i) To add delay in script
- ii) To terminate a process
- g) List any four operations performed on a file.

P.T.O.

- 2. Attempt any THREE of the following:** **12**
- a) Describe working of time sharing system with neat diagram.
 - b) List and describe any four services of operating system.
 - c) Differentiate between shared memory system and message passing system of interprocess communication.
 - d) State and describe any two scheduling criteria.
- 3. Attempt any THREE of the following:** **12**
- a) Describe use of ps and wait commands with suitable example.
 - b) Describe prevention of deadlock occurrence with respect to hold and wait necessary condition.
 - c) With suitable diagram, describe the concept of variable partitioning of memory.
 - d) Describe linked file allocation method with suitable example. Also list its one advantage.
- 4. Attempt any THREE of the following:** **12**
- a) Differentiate between command line based operating system and GUI based operating system (Any four points)
 - b) Describe various activities performed by following operating system components.
 - i) Main memory management
 - ii) File management
 - c) With suitable diagram, describe use of scheduling queues in process scheduling.
 - d) Write steps required for Banker's algorithm to avoid deadlock.
 - e) With suitable example, describe any one free space management technique.

5. Attempt any TWO of the following:**12**

- a) Calculate average waiting time for following data using First Come First Served (FCFS) and Shortest Job First (SJF) algorithms.

Process	Burst time	Arrival time
P0	08	0
P1	04	1
P2	05	2
P3	03	3

- b) Describe with example, use of following tools:
- Device Management
 - Performance monitor
 - Task scheduler
- c) Describe the concept of virtual memory with respect to paging. Also draw paging hardware diagram and describe its working with example.

6. Attempt any TWO of the following:**12**

- a) Describe one-to-one multithreading model with suitable diagram. Also write any two advantages of one-to-one model over many-to-one model.
- b) Consider the following page reference string arrival with three page frames:-
5, 6, 7, 8, 9, 7, 8, 5, 9, 7, 8, 7, 9, 6, 5, 6
Calculate number of page faults with optimal and FIFO (First In First Out) page replacement algorithms.
- c) Describe following directory structures in short with neat sketches:
- Single level
 - Two level
 - Tree structured



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Important Instructions to examiners:

- 1) The answers should be examined by key words and not as word-to-word as given in the model answer scheme.
- 2) The model answer and the answer written by candidate may vary but the examiner may try to assess the understanding level of the candidate.
- 3) The language errors such as grammatical, spelling errors should not be given more Importance (Not applicable for subject English and Communication Skills).
- 4) While assessing figures, examiner may give credit for principal components indicated in the figure. The figures drawn by candidate and model answer may vary. The examiner may give credit for anyequivalent figure drawn.
- 5) Credits may be given step wise for numerical problems. In some cases, the assumed constant values may vary and there may be some difference in the candidate's answers and model answer.
- 6) In case of some questions credit may be given by judgement on part of examiner of relevant answer based on candidate's understanding.
- 7) For programming language papers, credit may be given to any other program based on equivalent concept.
- 8) As per the policy decision of Maharashtra State Government, teaching in English/Marathi and Bilingual (English + Marathi) medium is introduced at first year of AICTE diploma Programme from academic year 2021-2022. Hence if the students in first year (first and second semesters) write answers in Marathi or bilingual language (English +Marathi), the Examiner shall consider the same and assess the answer based on matching of concepts with model answer.

Q. No	Sub Q.N.	Answer			Marking Scheme
1.	a) Ans.	Attempt any <u>FIVE</u> of the following: Differentiate between Multi programmed and Multitasking operating system (Any two points)			10 2M <i>Any two relevant points, 1M each</i>
		Features	Multiprogramming	Multitasking	
		Basic	It allows multiple programs to utilize the CPU simultaneously.	A supplementary of the multiprogramming system also allows for user interaction.	
		Mechanism	Based on the context switching mechanism.	Based on the time-sharing mechanism.	
		Objective	It is useful for reducing/decreasing	It is useful for running multiple processes at the	



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			CPU idle time and increasing throughput as much as possible.	same time, effectively increasing CPU and system throughput.	
		Execution	When one job or process completes its execution or switches to an I/O task in a multi-programmed system, the system momentarily suspends that process. It selects another process from the process scheduling pool (waiting queue) to run.	In a multiprocessing system, multiple processes can operate simultaneously by allocating the CPU for a fixed amount of time.	
		CPU Switching	In a multiuser environment, the CPU switches between programs/processes quickly.	In a single-user environment, the CPU switches between the processes of various programs.	
		Timing	It takes maximum time to execute the process.	It takes minimum time to execute the process.	
	b) Ans.	List any four services provided by O.S. - User Interface - Program Execution - I/O Operation - File system Manipulation - Communication - Error Detection - Resource Allocation - Accounting - Protection and security			2M <i>½ M each for any 4 services</i>



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c) Ans.	Define : Process, PCB Process:-A process is a program in execution. Process is also called as job, task or unit of work. PCB:-Process Control Block is a data structure that contains information of the process related to it. The process control block is also known as a task control block, entry of the process table, etc.	2M <i>Correct Definition 1M each</i>																					
d) Ans.	Define CPU and I/O burst cycle. CPU burst cycle: It is a time period when process is busy with CPU. I/O burst cycle: It is a time period when process is busy in working with I/O resources.	2M <i>Correct Definition 1M each</i>																					
e) Ans.	Differentiate between paging and segmentation. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">Parameters</th><th style="text-align: center;">Paging</th><th style="text-align: center;">Segmentation</th></tr> </thead> <tbody> <tr> <td>Individual Memory</td><td>In Paging, we break a process address space into blocks known as pages.</td><td>In the case of Segmentation, we break a process address space into blocks known as sections/segments.</td></tr> <tr> <td>Memory Size</td><td>The pages are blocks of fixed size.</td><td>The sections/segments are blocks of varying sizes.</td></tr> <tr> <td>Accountability</td><td>The OS divides the available memory into individual pages.</td><td>The compiler mainly calculates the size of individual segments, their actual address as well as virtual address.</td></tr> <tr> <td>Speed</td><td>This technique is comparatively much faster in accessing memory.</td><td>This technique is comparatively much slower in accessing memory than Paging.</td></tr> <tr> <td>Size</td><td>The available memory determines the individual page sizes.</td><td>The user determines the individual segment sizes.</td></tr> <tr> <td>Fragmentation</td><td>The Paging technique may underutilize some of the pages- thus</td><td>The Segmentation technique may not use some of the memory</td></tr> </tbody> </table>	Parameters	Paging	Segmentation	Individual Memory	In Paging, we break a process address space into blocks known as pages.	In the case of Segmentation, we break a process address space into blocks known as sections/segments.	Memory Size	The pages are blocks of fixed size.	The sections/segments are blocks of varying sizes.	Accountability	The OS divides the available memory into individual pages.	The compiler mainly calculates the size of individual segments, their actual address as well as virtual address.	Speed	This technique is comparatively much faster in accessing memory.	This technique is comparatively much slower in accessing memory than Paging.	Size	The available memory determines the individual page sizes.	The user determines the individual segment sizes.	Fragmentation	The Paging technique may underutilize some of the pages- thus	The Segmentation technique may not use some of the memory	2M <i>Any two relevant differences – 1M each</i>
Parameters	Paging	Segmentation																					
Individual Memory	In Paging, we break a process address space into blocks known as pages.	In the case of Segmentation, we break a process address space into blocks known as sections/segments.																					
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			leading to internal fragmentation.	blocks at all. Thus, it may lead to external fragmentation.	
		Logical Address	A logical address divides into page offset and page number in the case of Paging.	A logical address divides into section offset and section number in the case of Segmentation.	
		Data Storage	In the case of Paging, the page table leads to the storage of the page data.	In the case of Segmentation, the segmentation table leads to the storage of the segmentation data.	
	f) Ans.	Write syntax of following commands- (i) Kill (ii) Sleep i) kill Syntax: kill Pid ii) sleep Syntax: sleep NUMBER[SUFFIX]... sleep OPTION			2M <i>1M for each correct syntax</i>
	g) Ans.	List any four file operations. • Creating a file • Writing a file: • Reading a file: • Repositioning within a file • Deleting a file • Appending new information to the end of the file • Renaming an existing file. • Creating copy of a file, copy file to another I/O device such as printer or display			2M <i>Any four operations ½ M each</i>
2.	a) Ans.	Attempt any <u>THREE</u> of the following: Explain Time sharing O.S. In time sharing system, the CPU executes multiple jobs by switching among them. The switches occur so frequently that the users can interact with each program while it is running. It includes an			12 4M



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		interactive computer system which provides direct communication between the user and the system. A time-sharing system allows many users to share the computer resources simultaneously. The time-sharing system provides the direct access to many users where CPU time is divided among all the users on scheduled basis. The operating system allocates a time slice to each user. When this time is expired, it passes control to the next user on the system. The time allowed is extremely small and the users are given the impression that each of them has their own CPU and they are the sole owner of the CPU. In this time slice each user gets attention of the CPU. The objective of time-sharing system is to minimize response time of process. Example: The concept of time-sharing system is shown in figure: In above figure, the user 5 is active but user 1, user 2, user 3, and user 4 are in waiting state whereas user 6 is in ready status.	<i>Relevant Explanation</i> 4M
	b) Ans.	Describe any two components of O.S. List of System Components: 1. Process management 2. Main memory management 3. File management 4. I/O system management 5. Secondary storage management 1.Process Management: A program is a set of instructions. When CPU is allocated to a program, it can start its execution. A program in execution is a process. A word processing program run by a user on a PC is a process. A process needs various system resources including CPU time, memory, files and I/O devices to complete the job execution. These resources can be given to the process when it is created or allocated to it while it is running.	4M <i>Description of any two components of OS</i> 2M each



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	The operating system responsible for the following activities in connection with process management: • Creation and deletion of user and system processes. • Suspension and resumption of processes. • A mechanism for process synchronization. • A mechanism for process communication. • A mechanism for deadlock handling. 2. Main-Memory Management Main memory is a large array of words or bytes, ranging in size from hundreds of thousands to billions. Each word or byte has its own address. Main memory is a repository of quickly accessible data shared by the CPU and I/O devices. The central processor reads instructions from main memory during the instruction fetch cycle and both reads and writes data from main memory during the data fetch cycle. The main memory is generally the only large storage device that the CPU is able to address and access directly. The operating system responsible for the following activities in connection with main memory's management: • Keeping track of which parts of memory are currently being used and by whom. • Deciding which processes (or parts thereof) and data to move into and out of memory. 3. File Management A file is a collected of related information defined by its creator. Computer can store files on the disk (secondary storage), which provide long term storage. Some examples of storage media are magnetic tape, magnetic disk and optical disk. Each of these media has its own properties like speed, capacity, and data transfer rate and access methods. A file system normally organized into directories to ease their use. These directories may contain files and other directions. The operating system responsible for the following activities in connection with file management: • The creation and deletion of files. • The creation and deletion of directions. • The support of primitives for manipulating files and directions. • The mapping of files onto secondary storage. • The backup of files on stable storage media.	
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		4. I/O device Management Input / Output device management provides an environment for the better interaction between system and the I / O devices (such as printers, scanners, tape drives etc.). To interact with I/O devices in an effective manner, the operating system uses some special programs known as device driver. The device drivers take the data that operating system has defined as a file and then translate them into streams of bits or a series of laser pulses (in regard with laser printer). The I/O subsystem consists of several components: • A memory management component that includes buffering, caching, spooling • A general device driver interface • Drivers for specific hardware devices 5. Secondary-Storage Management The computer system provides secondary storage to back up main memory. Secondary storage is required because main memory is too small to accommodate all data and programs, and the data that it holds is lost when power is lost. Most of the programs including compilers, assemblers, word processors, editors, and formatters are stored on a disk until loaded into memory. Secondary storage consists of tapes drives, disk drives, and other media. The operating system is responsible for the following activities in connection with disk management: • Free space management • Storage allocation • Disk scheduling.	
	c) Ans.	Explain shared memory model of Interprocess communication (IPC) Inter-process communication: Cooperating processes require an Inter- process communication (IPC) mechanism that will allow them to exchange data and information.	4M <i>Explanation 3M Diagram 1M</i>



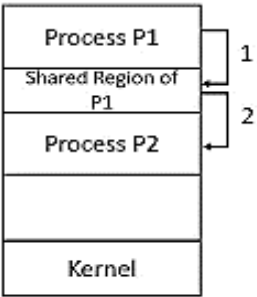
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		Shared memory  Shared Memory System • In this, all processes who want to communicate with other processes can access a region of the memory residing in an address space of a process creating a shared memory segment. • All the processes using the shared memory segment should attach to the address space of the shared memory. All the processes can exchange information by reading and/or writing data in shared memory segment. • The form of data and location are determined by these processes who want to communicate with each other. • These processes are not under the control of the operating system. • The processes are also responsible for ensuring that they are not writing to the same location simultaneously. • After establishing shared memory segment, all accesses to the shared memory segment are treated as routine memory access and without assistance of kernel.	
	d) Ans.	Describe different scheduling criteria. • CPU utilization: - In multiprogramming the main objective is to keep CPU as busy as possible. CPU utilization can range from 0 to 100 percent. • Throughput: - It is the number of processes that are completed per unit time. It is a measure of work done in the system. When CPU is busy in executing processes, then work is being done in the system. Throughput depends on the execution time required for any process.	4M <i>Any four scheduling criteria -1M each</i>



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		• Turnaround time: -The time interval from the time of submission of a process to the time of completion of that process is called as turnaround time. It is the sum of time period spent waiting to get into the memory, waiting in the ready queue, executing with the CPU, and doing I/O operations. • Waiting time: - It is the sum of time periods spent in the ready queue by a process. When a process is selected from job pool, it is loaded into the main memory. A process waits in ready queue till CPU is allocated to it.	
3.	a) Ans.	Attempt any <u>THREE</u> of the following: Draw and explain process state diagram. Different process states are as follows: 1. New 2. Ready 3. Running 4. Waiting 5. Terminated <pre> graph TD new((new)) -- admitted --> ready((ready)) ready -- scheduler dispatch --> running((running)) running -- interrupt --> ready running -- I/O or event wait --> waiting((waiting)) waiting -- I/O or event completion --> ready running -- exit --> terminated((terminated)) </pre> New: When a process enters into the system, it is in new state. In this state a process is created. In new state the process is in job pool. Ready: When the process is loaded into the main memory, it is ready for execution. In this state the process is waiting for processor allocation.	12 4M <i>Process state diagram</i> 2M <i>Explanation</i> 2M



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		Running: When CPU is available, system selects one process from main memory and executes all the instructions from that process. So, when a process is in execution, it is in running state. In single user system, only one process can be in the running state. In multiuser system, there can be multiple processes which are in the running state. Waiting State: When a process is in execution, it may request for I/O resources. If the resource is not available, process goes into the waiting state. When the resource is available, the process goes back to ready state. Terminated State: When the process completes its execution, it goes into the terminated state. In this state the memory occupied by the process is released.	
	b) Ans.	Describe conditions for deadlock prevention. By ensuring that at least one of below conditions cannot hold, we can prevent the occurrence of a deadlock. 1.Mutual Exclusion: The mutual-exclusion condition must hold for non-sharable resources. Sharable resources do not require mutually exclusive access, thus cannot be involved in a deadlock. 2.Hold and Wait: One way to avoid this Hold and Wait is when a process requests a resource; it does not hold any other resources. •One protocol that can be used requires each process to request and be allocated all its resources before it begins execution. •Another protocol that can be used is, to allow a process to request resources only when the process has none. A process may request some resources and use them. Before it requests any additional resources, it must release all the resources that are currently allocated to it. 3.No Preemption: If a process that is holding some resources requests another resource that cannot be immediately allocated to it, then all resources currently being held are preempted. That is these resources are implicitly	4M <i>Any four conditions 1M each</i>



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		released. The preempted resources are added to the list of resources for which the process is waiting. Process will be restarted only when all the resources i.e. its old resources, as well as the new ones that it is requesting will be available. 4.Circular Wait Circular-wait condition never holds is to impose a total ordering of all resource types, and to require that each process requests resources in an increasing order of enumeration. Let $R = \{R_1, R_2, \dots, R_n\}$ be the set of resource types. We assign to each resource type a unique integer number, which allows us to compare two resources and to determine whether one precedes another in our ordering. Formally, define a one-to-one function $F: R \rightarrow \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers.	
	c) Ans.	Explain fixed size memory partitioning. Fixed Size Memory Partitioning (Static) • Memory is divided into number of fixed size partitions, which is called as fixed or static memory partitioning. • Each partition contains exactly one process. • The number of programs to be executed depends on number of partitions. • When the partition is free, a selected process from the input queue is loaded into the free partition. • When the process terminates, the partition becomes available for another process. • The operating system keeps a table indicating parts of memory which are available and which are occupied. • Initially, all memory is available for user processes and it is considered as one large block of available memory, a hole. • When a process arrives, large enough hole of memory is allocated to the processes. •	4M <i>Correct Explanation 4M</i>
	d) Ans.	Explain linked file allocation method. Linked Allocation: • This allocation is on the basis of an individual block. Each block contains a pointer to the next block in the chain. • The disk block can be scattered anywhere on the disk.	4M <i>Correct explanation 4M,</i>



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		• The directory contains a pointer to the first and the last blocks of the file. • To create a new file, simply create a new entry in the directory. • The following figure shows the linked allocation. Directory <table><tr><th>file</th><th>start</th><th>end</th></tr><tr><td>jeep</td><td>9</td><td>25</td></tr></table><pre>graph TD subgraph Disk [Disk Blocks 0-31] 1 --> 10 10 --> 2 2 --> 26 26 --> 17 17 --> 16 16 --> 1 end subgraph Directory jeep["jeep start: 9 end: 25"] end jeep --> 9 jeep --> 25</pre> • There is no external fragmentation since only one block is needed at a time. • The size of a file need not be declared when it is created. • A file can continue to grow as long as free blocks are available • This method is used only for a sequential access files • This method requires more space to store pointers • So instead of blocks, clusters are used for allocation but this creates internal fragmentation.	file	start	end	jeep	9	25	<i>Diagram Optional</i>
file	start	end							
jeep	9	25							
4.	a)	Attempt any <u>THREE</u> of the following: Compare between command line and Graphical user interface. (Any four points)	12 4M						



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	Ans.	Parameter	Command Line Interface(CLI)	Graphic User Interface(GUI)	<i>Any four points 1M each</i>
		Definition	Interaction is by typing commands	Interaction with devices is by graphics and visual components and icons	
		Understanding	Commands need to be memorized	Visual indicators and icons are easy to understand	
		Memory	Less memory is required for storage	More memory is required as visual components are involved.	
		Working Speed	Use of keyboard for commands makes CLI quicker.	Use of mouse for interaction makes it slow	
		Resources used	Only keyboard	Mouse and keyboard both can be used	
		Accuracy	High	Comparatively low	
		Flexibility	Command line interface does not change, remains same over time	Structure and design can change with updates	
	b) Ans.	Write any four systems call related to file management. System calls related to file management are: 1. create new file 2. delete existing file 3. open file 4. close file 5. create directories 6. delete directories 7. read, write, reposition in file 8. getfile attributes 9. set file attributes			4M <i>Any 4 system calls 1M each</i>



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c) Ans.	Compare between Long term and short term scheduler. (Any four points)			4M <i>Any four points 1M each</i>
	Sr. No	Long Term Scheduler	Short Term Scheduler	
	1	It is job scheduler	It is CPU scheduler	
	2	It selects processes from job pool and loads them into memory for execution	It selects processes from ready queue which are ready to execute and allocates CPU to one of them	
	3	Access job pool and ready queue	Access ready queue and CPU	
	4	It executes much less frequently. It executes when memory has space to accommodate new process.	It executes frequently. It executes when CPU is available for allocation	
	5	Speed is less than short term scheduler	Speed is fast	
	6	It controls the degree of multiprogramming	It provides lesser control over degree of multiprogramming	
	7	It chooses a good process that is a mix-up of input/output bound and CPU bound.	It chooses a new process for a processor quite frequently.	
d)	Solve given problem by using SJF and FCFS scheduling algorithm using Gantt chart. Calculate the average waiting time for each algorithm			4M



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Process	Burst time (in ms)
P1	9
P2	7
P3	3
P4	7

Ans. Gantt Chart SJF

P3	P2	P4	P1
0	3	10	17
26			

Waiting Time
P1=17
P2=3
P3=0
P4=10
Average waiting time=Waiting time of all processes / Number of processes

$$= (17+3+0+10) / 4$$

$$= 30/4$$

$$= 7.5 \text{ milliseconds (ms)}$$

Gantt Chart FCFS

P1	P2	P3	P4
0	9	16	19
26			

Waiting Time
P1=0
P2=9
P3=16
P4=19
Average waiting time=Waiting time of all processes / Number of processes

$$= (0+9+16+19) / 4$$

$$= 44/4$$

$$= 11 \text{ milli seconds (ms)}$$

For each scheduling Gantt chart 1M,

Each average waiting time calculation 1M

For each scheduling Gantt chart 1M,

Each average waiting time calculation 1M



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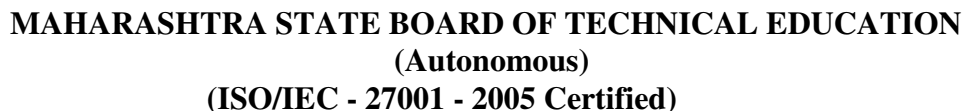
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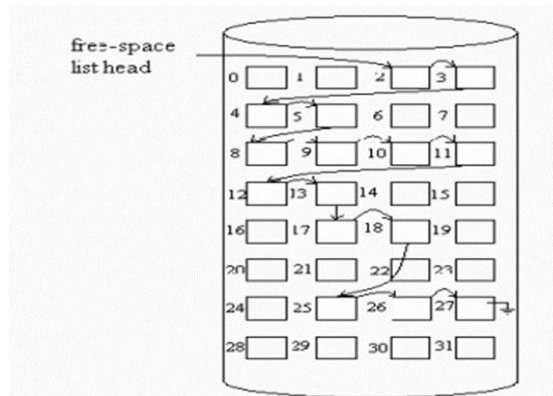
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e) Ans.	Describe free space management technique. (Any two) A file system is responsible to allocate the free blocks to the file therefore it has to keep track of all the free blocks present in the disk. There are mainly four approaches by using which, the free blocks in the disk are managed. 1. Bit Vector 2. Linked List 1)Bit Vector: The free-space list is implemented as a bit map or bit vector. Each block is represented by 1 bit. If the block is free, the bit is 1; if the block is allocated, the bit is 0. For example, consider a disk where blocks 2, 3, 4, 5, 8, 9, 10, 11, 12, 13 are free and the rest of the blocks are allocated. The free-space bit map would be : 0011110011111100 <table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> 1=Free block 0= Allocated block The main advantage of this approach is its relative simplicity and its efficiency in finding the first free block or n consecutive free blocks on the disk. 2) Linked List In this approach, the free disk blocks are linked together i.e. a free block contains a pointer to the next free block. The block number of the very first disk block is stored at a separate location on disk and is also cached in memory. In this approach, link all the disk blocks together, keeping a pointer to the first free block. This block contains a pointer to the next free disk block, and so on.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0	4M <i>Any 2 techniques Correct Explanation 2M each</i>
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15																			
0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0																			



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5.	a) Attempt any <u>TWO</u> of the following: Write two uses of following O.S. tools (i) Device Management (ii) Performance monitor (iii) Task Scheduler Ans. i) Device management: • Managing all the hardware or virtual devices of computer system. • Allow interaction with hardware devices through device driver. • Used to install device and component-level drivers as well as associated software. • Allocate devices to the process as per process requirement and priority. • Deallocate devices either temporarily or permanently depending on condition. • Keeping track of all device's data and location. • Monitoring device status like printers, storage drivers and other devices. • Used to enforce the predetermined policies and decides which process receives the device when and for how long. ii) Performance monitor 1. Monitor various activities on a computer such as CPU or memory usage. 2. Used to examine how programs running on their computer affect computer's performance 3. It is used to identify performance problems or bottleneck that affect operating system or installed applications. 4. Used to observe the effect of system configuration changes.	12 6M <i>2 uses of each tool 2M</i>
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		iii) Task scheduler 1. Assign processor to task ready for execution 2. Executing predefined actions automatically whenever a certain set of condition is met. <i>(Any two relevant uses shall be considered)</i>																																																																																																																																																																																																																			
	b) Ans.	Writer the outputs of following commands (i) Wait 2385018 (ii) Sleep 09 (iii) PS -u Asha i) Wait command waits until the termination of specified process ID 2385018 ii) Sleep command is used to delay for 9 seconds during the execution of a process i.e. it will pause the terminal for 9 seconds. iii) ps command with -u is used to display data/processes for the specific user Asha.	6M <i>2M for each correct output</i>																																																																																																																																																																																																																		
	c) Ans.	Given a page reference string with three (03) page frames. Calculate the page faults with ‘Optimal’ and ‘LRU’ page replacement algorithm respectively. ‘7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1’ (Representation of frame can be in any order) i) Optimal <table border="1"><tr><td>Ref</td><td>7</td><td>0</td><td>1</td><td>2</td><td>0</td><td>3</td><td>0</td><td>4</td><td>2</td><td>3</td><td>0</td><td>3</td><td>2</td><td>1</td><td>2</td><td>0</td><td>1</td><td>7</td><td>0</td><td>1</td></tr><tr><td>F1</td><td>7</td><td>7</td><td>7</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>7</td><td>7</td><td>7</td></tr><tr><td>F2</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>4</td><td>4</td><td>4</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>F3</td><td></td><td></td><td>1</td><td>1</td><td>1</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Fault</td><td>F</td><td>F</td><td>F</td><td>F</td><td></td><td>F</td><td></td><td>F</td><td></td><td></td><td>F</td><td></td><td></td><td>F</td><td></td><td></td><td></td><td>F</td><td></td><td></td></tr></table> Total page faults- 9 ii) LRU <table border="1"><tr><td>Ref</td><td>7</td><td>0</td><td>1</td><td>2</td><td>0</td><td>3</td><td>0</td><td>4</td><td>2</td><td>3</td><td>0</td><td>3</td><td>2</td><td>1</td><td>2</td><td>0</td><td>1</td><td>7</td><td>0</td><td>1</td></tr><tr><td>F1</td><td>7</td><td>7</td><td>7</td><td>2</td><td>2</td><td>2</td><td>2</td><td>4</td><td>4</td><td>4</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>F2</td><td></td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>F3</td><td></td><td></td><td>1</td><td>1</td><td>1</td><td>3</td><td>3</td><td>3</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>2</td><td>7</td><td>7</td><td>7</td></tr><tr><td>Fault</td><td>F</td><td>F</td><td>F</td><td>F</td><td></td><td>F</td><td></td><td>F</td><td>F</td><td>F</td><td>F</td><td></td><td></td><td>F</td><td></td><td>F</td><td></td><td>F</td><td></td><td></td></tr></table> Total page faults-12	Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1	F1	7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7	7	7	F2		0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0	0	0	F3			1	1	1	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1	Fault	F	F	F	F		F		F			F			F				F			Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1	F1	7	7	7	2	2	2	2	4	4	4	0	0	0	1	1	1	1	1	1	1	F2		0	0	0	0	0	0	0	0	3	3	3	3	3	3	0	0	0	0	0	F3			1	1	1	3	3	3	2	2	2	2	2	2	2	2	2	7	7	7	Fault	F	F	F	F		F		F	F	F	F			F		F		F			6M <i>Calculate page fault with relevant diagram- 3M each</i>
Ref	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1																																																																																																																																																																																																	
F1	7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7	7	7																																																																																																																																																																																																	
F2		0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0	0	0																																																																																																																																																																																																	
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6.

a)

Attempt any **TWO** of the following:

Solve given problem by using

(i) Pre-emptive SJF

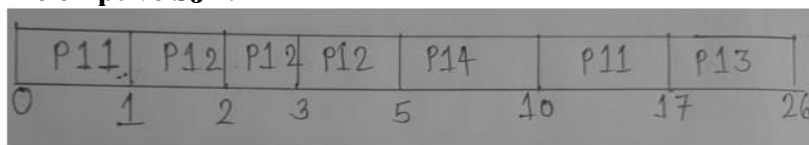
(ii) Round Robin (Time Slice = 3 ms)

Calculate average waiting time using Gantt Chart

Process	A.T.	B.T. (in ms)
P ₁₁	0	8
P ₁₂	1	4
P ₁₃	2	9
P ₁₄	3	5

Ans.

(i) Pre-emptive SJF:



Waiting Time= (Total completion time –Burst time) – Arrival time

$$P_{11} - (17 - 8) - 0 = 9\text{ms},$$

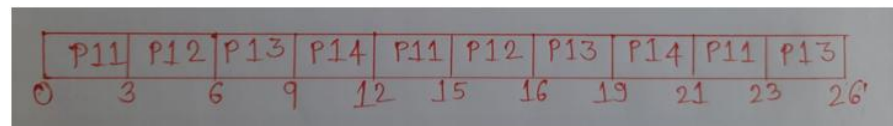
$$P_{12} - (5 - 4) - 1 = 0\text{ms},$$

$$P_{13} - (26 - 9) - 2 = 15\text{ms},$$

$$P_{14} - (10 - 5) - 3 = 2\text{ms}$$

$$\text{Average waiting time :- } (9 + 0 + 15 + 2) / 4 = 26 / 4 = 6.5 \text{ ms}$$

(ii) Round Robin (Time Slice = 3 ms)



$$\text{Waiting time: - } P_{11} = (23 - 8) - 0 = 15\text{ms},$$

$$P_{12} - (16 - 4) - 1 = 11\text{ms},$$

$$P_{13} - (26 - 9) - 2 = 15\text{ms},$$

$$P_{14} - (21 - 5) - 3 = 13\text{ms}$$

$$\text{Average waiting time:- } (15 + 11 + 15 + 13) / 4 = 54 / 4 = 13.5\text{ms}$$

12
6M

Each
method 3M
-
1M for
Gantt chart,

1M for
Waiting
time
calculation,

1M for
Average
waiting time



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	b) Consider the following memory map and assume a new process P4 comes with memory requirements of 6 KB. Locate (Draw) this process in memory using. i) First fit ii) Best Fit iii) Worst Fit <table><tr><td>O.S.</td></tr><tr><td>P1</td></tr><tr><td><FREE> 12 KB</td></tr><tr><td>P2</td></tr><tr><td><FREE> 19 KB</td></tr><tr><td>P3</td></tr><tr><td><FREE> 7KB</td></tr><tr><td>Memory</td></tr></table>	O.S.	P1	<FREE> 12 KB	P2	<FREE> 19 KB	P3	<FREE> 7KB	Memory	6M Each fit diagram 2M																
O.S.																										
P1																										
<FREE> 12 KB																										
P2																										
<FREE> 19 KB																										
P3																										
<FREE> 7KB																										
Memory																										
Ans.	First Fit: Allocate the first free block to the new process P4. <table><tr><td>O. S.</td></tr><tr><td>P1</td></tr><tr><td>P4 6KB</td></tr><tr><td><FREE> 6KB</td></tr><tr><td>P2</td></tr><tr><td><FREE> 19 KB</td></tr><tr><td>P3</td></tr><tr><td><FREE> 7 KB</td></tr></table> Best Fit: Allocate the smallest free block that is big enough to accommodate new process P4. <table><tr><td>O. S.</td></tr><tr><td>P1</td></tr><tr><td><FREE> 12 KB</td></tr><tr><td>P2</td></tr><tr><td><FREE> 19 KB</td></tr><tr><td>P3</td></tr><tr><td>P4 6 KB</td></tr><tr><td><FREE> 1 KB</td></tr></table> Worst fit: Allocate the largest free block to the new process P4. <table><tr><td>O. S.</td></tr><tr><td>P1</td></tr><tr><td><FREE> 12 KB</td></tr><tr><td>P2</td></tr><tr><td>P4 6 KB</td></tr><tr><td><FREE> 13 KB</td></tr><tr><td>P3</td></tr><tr><td><FREE> 7 KB</td></tr></table>	O. S.	P1	P4 6KB	<FREE> 6KB	P2	<FREE> 19 KB	P3	<FREE> 7 KB	O. S.	P1	<FREE> 12 KB	P2	<FREE> 19 KB	P3	P4 6 KB	<FREE> 1 KB	O. S.	P1	<FREE> 12 KB	P2	P4 6 KB	<FREE> 13 KB	P3	<FREE> 7 KB	
O. S.																										
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P2																										
P4 6 KB																										
<FREE> 13 KB																										
P3																										
<FREE> 7 KB																										
c) Ans.	Construct and explain directory structure of a file system in terms of two level and tree structure. 1) Two-level directory: - In the two-level structures, each user has its own user file directory (UFD). The UFD lists only files of a single user. System contains a master file directory (MFD) which is indexed by user name or account number. Each entry in MFD points to the UFD for that user. When a user refers to a particular file, only his own UFD is searched. Different users can have files with the same name, as long as all the file names within each UFD are unique.	6M Explanation of structure 2M each, Construction of structure 1M each																								

6M

Each fit
diagram 2M

6M

Explanation
of structure
2M each,

Constructio
n of
structure
1M each



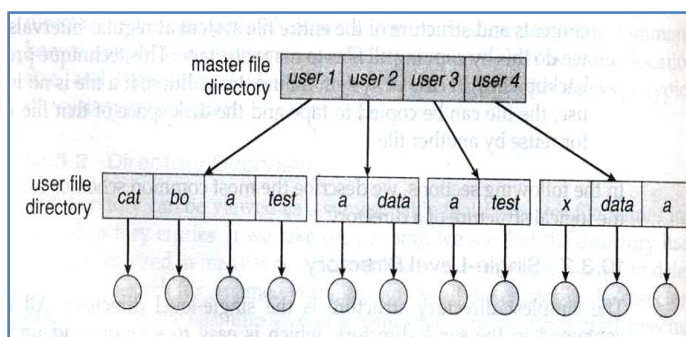
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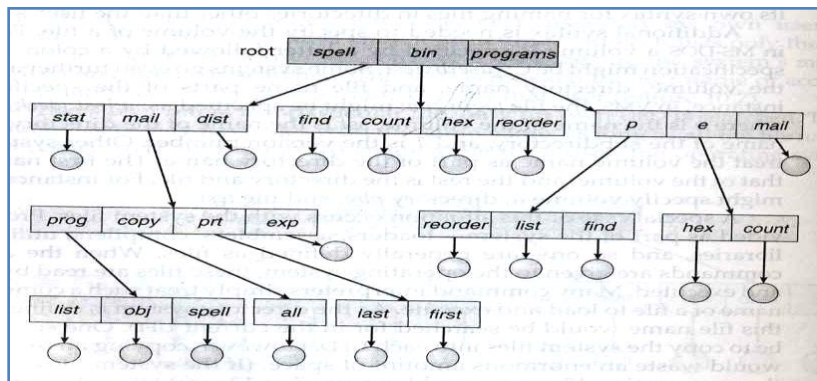
22516

When we create a file for a user, operating system searches only that user's UFD to find whether same name file already present in the directory. For deleting a file again operating system checks the file name in the user's UFD only.



2. Tree structure:-

In this directory structure user can create their own sub-directories and organize their files. The tree has a root directory and every file has a unique path name. A directory contains a set of files or subdirectories. All directories have the same internal format. One bit in each directory entry defines the entry as a file (0) or as a subdirectory (1). Each process has a current directory. Current directory contains files that are currently required by the process. When reference is made to a file, the current directory is searched. If a file needed that is not in the current directory, then the user usually must either specify a path name or change the current directory.



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22223

3 Hours / 70 Marks

Seat No.

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- Instructions* –
- (1) All Questions are *Compulsory*.
 - (2) Illustrate your answers with neat sketches wherever necessary.
 - (3) Figures to the right indicate full marks.
 - (4) Assume suitable data, if necessary.
 - (5) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.
 - (6) Preferably, write the answer in sequential order.

Marks

- 1. Attempt any FIVE of the following:** **10**
- a) State any two features of Linux.
 - b) Difference between Time sharing system and Real time system (any 2 points)
 - c) State any four services of operating system.
 - d) Write the difference between pre-emptive and non-preemptive scheduling.
 - e) Define following terms :
 - i) Virtual Memory
 - ii) Paging
 - f) List any four file attributes and it's meaning.
 - g) Define Deadlock.

P.T.O.

- 2. Attempt any THREE of the following:** **12**
- a) Explain different types of system calls.
 - b) Draw and explain process control block in detail.
 - c) State and explain four scheduling criteria.
 - d) Define fragmentation. Explain Internal and External Fragmentation
- 3. Attempt any THREE of the following:** **12**
- a) Describe multiprocessor OS with it's advantages. (any two)
 - b) Explain different components of operating system.
 - c) Explain different types of schedulers.
 - d) Explain two level directory structure with suitable diagram.
- 4. Attempt any THREE of the following:** **12**
- a) Explain real time OS. Explain it's types.
 - b) Draw process state diagram and describe each state.
 - c) Describe any four condition for deadlock.
 - d) Differentiate between paging and segmentation (any 4 points)
 - e) Explain fixed and variable memory management.
- 5. Attempt any TWO of the following:** **12**
- a) Explain the working of interprocess communication considering
 - i) Shared memory
 - ii) Message passing
 - b) With neat diagram explain multilevel queue scheduling.
 - c) Write two uses of following operating system tools.
 - i) Security policy
 - ii) User Management
 - iii) Task scheduler

6. Attempt any TWO of the following:**12**

a) List file allocation method and explain any one in details.

b) For the page reference string

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

Calculate the page faults applying.

i) Optimal

ii) LRU

iii) FiFo page

Replacement algorithms for a memory with three frames.

c) Consider the four processes P1, P2, P3 and P4 with length of CPO Burst time. Find out Avg waiting time and Avg turn around time for the following Algorithms.

i) FCFS

ii) RR (Slice-4ms)

iii) SJF

Process	Arrival time	Burst time
P1	0	8
P2	1	4
P3	2	9
P4	3	5



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Important Instructions to examiners:

- 1) The answers should be examined by key words and not as word-to-word as given in the model answer scheme.
- 2) The model answer and the answer written by candidate may vary but the examiner may try to assess the understanding level of the candidate.
- 3) The language errors such as grammatical, spelling errors should not be given more Importance (Not applicable for subject English and Communication Skills).
- 4) While assessing figures, examiner may give credit for principal components indicated in the figure. The figures drawn by candidate and model answer may vary. The examiner may give credit for any equivalent figure drawn.
- 5) Credits may be given step wise for numerical problems. In some cases, the assumed constant values may vary and there may be some difference in the candidate's answers and model answer.
- 6) In case of some questions credit may be given by judgement on part of examiner of relevant answer based on candidate's understanding.
- 7) For programming language papers, credit may be given to any other program based on equivalent concept.
- 8) As per the policy decision of Maharashtra State Government, teaching in English/Marathi and Bilingual (English + Marathi) medium is introduced at first year of AICTE diploma Programme from academic year 2021-2022. Hence if the students in first year (first and second semesters) write answers in Marathi or bilingual language (English + Marathi), the Examiner shall consider the same and assess the answer based on matching of concepts with model answer.

Q. No	Sub Q.N.	Answer	Marking Scheme
1.	a) Ans.	Attempt any <u>FIVE</u> of the following: Define real time operating system, along with any two applications of it. Real Time operating system is a special-purpose operating system used in computers that has well defined fixed time constraints for processing any task. OR Real-time operating systems (RTOS) are used in environments where a large number of events, mostly external to the computer system, must be accepted and processed in a short time or within certain deadlines. OR Any relevant Definition Applications: (any other relevant application shall be considered)	10 2M <i>Definition</i> <i>1M,</i> <i>Any two</i> <i>application</i> <i>s</i> <i>1/2M each</i>



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		1. Flight Control System 2. Simulations 3. Industrial control 4. Military applications 5. Airline Traffic Control System 6. Airline Reservation System 7. Defence application systems like RADAR 8. Heart Pacemaker 9. Network Multimedia Systems 10. Missile Guidance system 11. Weather Forecasting 12. Online Transaction system 13. Network and Multimedia Systems 14. Ticket Reservation System 15. Command Control Systems 16. Medical Critical Care Systems.	
	b) Ans.	List any four services provided by operating system. • User Interface • Program Execution • I/O Operation • File system Manipulation • Communication • Error Detection • Resource Allocation • Accounting • Protection and security	2M <i>Any four 1/2M each</i>
	c)	Draw neat labelled process state diagram along with the correct directions of arrows	2M



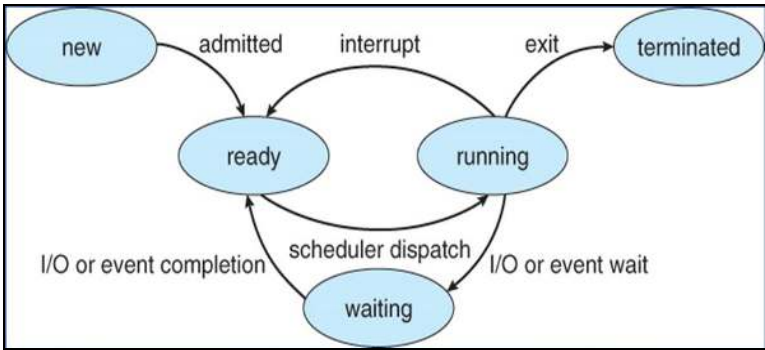
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	Ans.	 <pre> graph LR new([new]) -- admitted --> ready([ready]) ready -- scheduler dispatch --> running([running]) running -- interrupt --> ready running -- "I/O or event wait" --> waiting([waiting]) waiting -- "I/O or event completion" --> ready running -- exit --> terminated([terminated]) </pre>	Correct diagram 2M
	d) Ans.	Define CPU bound program and I/O bound program CPU bound program: If execution of a program is highly dependent on the CPU then it is known as CPU bound program. OR If a task does a lot operations using CPU, it's called a CPU-bound task. I/O bound program: If execution of a program is dependent on the input-output system and its resources, such as disk drives and peripheral devices then it is known as I/O bound program.	2M Each definition 1M
	e) Ans.	Define paging and segmentation Paging is a memory management scheme that permits the physical address space of process to be noncontiguous. Logical memory is divided into blocks of same size called as pages. OR Paging is a storage mechanism used in OS to retrieve processes from secondary storage to the main memory as pages. OR The process of retrieving processes in the form of pages from the secondary storage into the main memory is known as paging. Segmentation is a memory management scheme that permits dividing logical address space into multiple segments. OR Segmentation divides the Computer's physical memory and program's address space into segments.	2M Correct definition 1M each



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	f) Ans.	What is the use of PS command? Write long forms of UID, PID in the output of this command. ps command: It is used to display the characteristics of a process. This command is used to list the processes associated with a user at a particular terminal. UID - (Owner) User-id PPID - Parent Process-id	2M <i>Use of ps command 1M</i> <i>full form of each 1/2M</i>
	g) Ans.	List any four file operations. • Creating a file • Writing a file • Reading a file • Repositioning within a file • Deleting a file • Appending new information to the end of the file • Renaming an existing file. • Truncating a file • Creating copy of a file, copy file to another I/O device such as printer or display	2M <i>Any four operations 1/2M each</i>
2.	a) Ans.	Attempt any <u>THREE</u> of the following: Describe multiprocessor OS with its advantages (any two) • Multiprocessor systems are also known as parallel systems or tightly coupled systems. • These systems have two or more processors in close communication and they share computer resources such as bus, clock, memory and peripheral devices. • The whole task of multiprocessing is managed by the operating system, which allocates different tasks to be performed by the various processors in the system. • Applications designed for the use in multiprocessing are said to be threaded, which means that they are broken into smaller routines that can be run independently. • Multiple CPUs are linked together so that a job can be divided and executed more quickly. When a job is completed, the results from all CPUs are compiled to provide the final output.	12 4M <i>Description of multiprocessor OS- 3M</i> <i>Any two advantages 1/2M each</i>



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		OR Note: Any other relevant advantage shall be considered Advantages • Increased reliability: Due to the multiprocessing system, processing tasks can be distributed among several processors. This increases reliability as if one processor fails; the task can be given to another processor for completion. • Increased throughput: As several processors increase, more work can be done in less time. • The economy of Scale: As multiprocessors systems share peripherals, secondary storage devices, and power supplies, they are relatively cheaper than single-processor systems. • Reduce Cost	
	b) Ans.	Write down the responsibilities of the following components of OS i) Memory Management ii) File Management i) Memory Management: • Keeping track of which parts of memory are currently being used and by whom.	4M <i>Responsibilities of each management</i> 2M



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		- Deciding which processes (or parts thereof) and data to move into and out of memory. - Allocating and deallocating memory space as needed. - Managing swap spaces, which store inactive pages of memory. - Implementing policies for memory allocation. File Management: - The creation and deletion of files. - The creation and deletion of directory - The support of primitives for manipulating files and directions. - The mapping of files onto secondary storage. - The backup of files on stable storage media. - Adding and editing the data in files. - Moving files from one location to another. - Store, arrange, or accessing files on a disk or other storage locations.	
	c) Ans.	Explain shared memory method of IPC using neat labelled diagram Inter-process communication: Cooperating processes require an Inter- process communication (IPC) mechanism that will allow them to exchange data and information. <pre> graph TD P1[Process P1] -- 1 --> SR[Shared Region of P1] SR -- 2 --> P2[Process P2] K[Kernel] style P1 fill:#fff,stroke:#000 style SR fill:#fff,stroke:#000 style P2 fill:#fff,stroke:#000 style K fill:#fff,stroke:#000 </pre> Shared Memory System - In this, all processes who want to communicate with other processes can access a region of the memory residing in an address space of a process creating a shared memory segment. - All the processes using the shared memory segment should attach to the address space of the shared memory. All the processes can exchange information by reading and/or writing data in shared memory segment.	4M <i>Explanation 3M, diagram 1M</i>



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		• The form of data and location are determined by these processes who want to communicate with each other. • These processes are not under the control of the operating system. • The processes are also responsible for ensuring that they are not writing to the same location simultaneously. • After establishing shared memory segment, all accesses to the shared memory segment are treated as routine memory access and without assistance of kernel.	
	d) Ans.	Explain following terms with respect to scheduling i) CPU utilization ii) Throughput iii) Turnaround time iv) Waiting time • CPU utilization: In multiprogramming the main objective is to keep CPU as busy as possible. CPU utilization can range from 0 to 100 percent. • Throughput: It is the number of processes that are completed per unit time. It is a measure of work done in the system. Throughput depends on the execution time required for any process. • Turnaround time: The time interval from the time of submission of a process to the time of completion of that process is called as turnaround time. It is the sum of time period spent waiting to get into the memory, waiting in the ready queue, executing with the CPU, and doing I/O operations. • Waiting time: It is the sum of time periods spent in the ready queue by a process. When a process is selected from job pool, it is loaded into the main memory. A process waits in ready queue till CPU is allocated to it.	4M <i>Each term 1M</i>
3.	a)	Attempt any <u>THREE</u> of the following: Explain following commands with their syntax i) kill ii) Sleep iii) Wait iv) Exit	12 4M



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Ans.	i)kill Syntax: kill pid Kill command is used to stop execution of particular process by sending an interrupt signal to the process ii)Sleep Syntax: sleep NUMBER[SUFFIX] The sleep command pauses the execution for specified time in command. iii)Wait Syntax: wait [pid] Wait command waits for running process to complete and return the exit status. iv) Exit Syntax: exit used to quit the shell (OR) Syntax: exit[n] The terminal window will close and return a status of n	<i>Each command Syntax 1/2M</i> <i>Explanation 1/2 M</i>
b) Ans.	What is deadlock? Discuss any one method of deadlock prevention A deadlock is a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process. Methods for deadlock prevention (<i>Any 1</i>) 1. Eliminate Mutual Exclusion: The mutual-exclusion condition must hold for non-sharable resources. Mutual section from the resource point of view is the fact that a resource can never be used by more than one process simultaneously which is fair enough but that is the main reason behind the deadlock. If a resource could have been used by more than one process at the same time then the process would have never been waiting for any resource. Read-only files are a good example of a sharable resource. If several processes attempt to open a read-only file at the same time, they can be granted simultaneous access to the file.	4M <i>Deadlock definition 1M</i> <i>Any 1 method 3M</i>



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		A process never needs to wait for a sharable resource. Sharable resources, in contrast, do not require mutually exclusive access and thus cannot be involved in a deadlock 2. Eliminate Hold and Wait: One way to avoid this Hold and Wait is when a process requests a resource; it does not hold any other resources. One protocol that can be used requires each process to request and be allocated all its resources before it begins execution. Another protocol that can be used is, to allow a process to request resources only when the process has none. A process may request some resources and use them. Before it requests any additional resources, it must release all the resources that are currently allocated to it. 3. Eliminate No Preemption: If a process that is holding some resources requests another resource that cannot be immediately allocated to it, then all resources currently being held are preempted. That is these resources are implicitly released. The preempted resources are added to the list of resources for which the process is waiting. Process will be restarted only when all the resources i.e. its old resources, as well as the new ones that it is requesting will be available. Preemption ensures that resources are efficiently utilized and prevents deadlocks caused by the hold and wait condition. 4. Eliminate Circular Wait: Circular-wait condition never holds is to impose a total ordering of all resource types, and to require that each process requests resources in an increasing order of enumeration. Let $R = \{R_1, R_2, \dots, R_n\}$ be the set of resource types. We assign to each resource type a unique integer number, which allows us to compare two resources and to determine whether one precedes another in our ordering. Formally, define a one-to-one function $F: R \rightarrow \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers.	
	c)	Describe concept of free space management technique using bit map method	4M



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	Ans. Bit Map: The free-space list is implemented as a bit map or bit vector. Each block is represented by 1 bit. If the block is free, the bit is 1; if the block is allocated, the bit is 0. For example, consider a disk where blocks 2, 3, 4, 5, 8, 9, 10, 11, 12, 13 are free and the rest of the blocks are allocated. The free-space bit map would be : 001111001111100 <table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> 1=Free block 0=Allocated block The main advantage of this approach is its relative simplicity and its efficiency in finding the first free block or n consecutive free blocks on the disk.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0	<i>Correct Explanation 4M</i>
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15																			
0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0																			
d) Ans.	Draw the diagram of linked file allocation method and explain it. Linked Allocation: This allocation is on the basis of an individual block. Each block contains a pointer to the next block in the chain. • In this scheme, each file is a linked list of disk blocks which need not be contiguous. The disk blocks can be scattered anywhere on the disk. • The directory contains a pointer to the first and the last blocks of the file. • Creating a file is easy. We simply create a new entry in the device directory. A write to a file removes first free block from free space list and write to it. This new block is then linked to the end of the file. To read a file, we simply read block by following the pointers from block to block. • There is no external fragmentation with linked allocation.	4M <i>Correct Diagram 2M</i> <i>Explanation 2M</i>																																



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		- Space is required for the pointers, 1.5% of disk is used for the pointers and not for information. If a pointer is lost or damaged or bug occurs in operating system or disk hardware failure occur, it may result in picking up the wrong pointer. - This method cannot support direct access. - This method is used only for a sequential access files - This method requires more space to store pointers - So instead of blocks, clusters are used for allocation but this creates internal fragmentation.	
4.	a)	Attempt any <u>THREE</u> of the following: Compare between CLI based OS and GUI based OS (Any four points)	12 4M



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Ans.		COMMAND LINE INTERFACE (CLI)	GRAPHIC USER INTERFACE (GUI)	Any 4 points 1M each
	Definition	Interaction is by typing commands	Interaction with devices is by graphics and visual components and icons	
	Understanding	Commands need to be memorized	Visual indicators and icons are easy to understand	
	Memory	Less memory is required for storage	More memory is required as visual components are involved.	
	Working Speed	Use of keyboard for commands makes CLI quicker and faster	Use of mouse for interaction makes it slow	
	Resources used	Only keyboard	Mouse and keyboard both can be used	
	Accuracy	High	Comparatively low	
	Menu	In CLI, there are no menus provided.	While in GUI, menus are provided	
	Flexibility	Command line interface does not change, remains same over time	Structure and design can change with updates	
	Example	DOS, UNIX	Windows, Linux	



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b) Ans.	What are the different types of system calls? Give examples of each Types of System Calls: 1.Process Control 2.File management 3.Device Management 4.Information Maintenance 5.Communication: 1. System calls for Process Control: ● Create/ Terminate process ● Load/Execute Process. ● End/Abort process. ● Ready process/Dispatch process. ● Suspend process/Resume Process. ● Get/Set Process Attributes. ● Wait event, signal event ● Allocate and deallocate memory 2. System call for File management: ● create new file, delete existing file ● open, close file ● Create and delete directories ● read, write, reposition ● get/set file attributes 3. System call for Device Management: ● request device, release device ● read, write, reposition ● get/set device attributes ● logically attach or detach devices 4. System call for Information Maintenance: ● get/set time or date ● get/set system data ● get/set process, file, or device attributes 5. System Communication: ● create, delete communication connection ● send, receive messages ● transfer status information ● attach or detach remote devices	4M <i>Listing of Types 2M</i> <i>Example of any two system calls 2M</i>
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c)	Explain working of CPU switch from process to process with neat labelled diagram.	4M
Ans.	A CPU switch from process to process is referred as context switch. A context switch is a mechanism that store and restore the state or context of a CPU in Process Control block sothat a process execution can be resumed from the same point at a later time. When the scheduler switches the CPU from one process to another process, the context switch save the contents of all process registers for the process being removed from the CPU, in its process control block. Context switch includes two operations such as state save and state restore. State save operation stores the current information of running process into its PCB. State restore operation restores the information of process to be executed from its PCB. Switching the CPU from one process to another process requires performing state save operation for the currently executing process (blocked) and a state restore operation for the process ready for execution. This task is known as context switch. <pre>graph TD subgraph OS [Operating System] direction TB S0[Save state into PCB0] R1[Reload state from PCB1] S1[Save state into PCB1] R0[Reload state from PCB0] end P0[Process P0] P1[Process P1] P0 -- Executing --> S0 S0 --> P0 -- Idle P1 -- Idle --> R1 R1 --> P1 -- Executing P1 -- Executing --> S1 S1 --> P1 -- Idle P0 -- Idle --> R0 R0 --> P0 -- Executing</pre>	Explanation 2M Diagram 2M Relevant Explanation shall be considered



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d)	Solve given problem by using FCFS scheduling algorithm. Draw correct Gantt chart and calculate average waiting time and average turnaround time <table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr><tr><td>P0</td><td>0</td><td>10</td></tr><tr><td>P1</td><td>1</td><td>29</td></tr><tr><td>P2</td><td>2</td><td>3</td></tr><tr><td>P3</td><td>3</td><td>7</td></tr><tr><td>P4</td><td>4</td><td>12</td></tr></table> Gantt Chart <	Process	Arrival Time	Burst Time	P0	0	10	P1	1	29	P2	2	3	P3	3	7	P4	4	12
Process	Arrival Time	Burst Time																	
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e)	Which hole is taken for next segment request for 8 KB in a swapping system for First fit, Best fit and Worst fit. <table><tr><td>OS</td></tr><tr><td>4 KB</td></tr><tr><td>9 KB</td></tr><tr><td>20 KB</td></tr><tr><td>16 KB</td></tr><tr><td>8 KB</td></tr><tr><td>2 KB</td></tr><tr><td>6 KB</td></tr></table>	OS	4 KB	9 KB	20 KB	16 KB	8 KB	2 KB	6 KB	4M																																																			
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First Fit: Allocate the first free block to the new process <table><tr><td>OS</td></tr><tr><td>4 KB</td></tr><tr><td>9 KB</td></tr><tr><td>20 KB</td></tr><tr><td>16 KB</td></tr><tr><td>8 KB</td></tr><tr><td>2 KB</td></tr><tr><td>6 KB</td></tr></table>	OS	4 KB	9 KB	20 KB	16 KB	8 KB	2 KB	6 KB	Best Fit: Allocate the smallest free block that is big enough to accommodate new process <table><tr><td>OS</td></tr><tr><td>4 KB</td></tr><tr><td>9 KB</td></tr><tr><td>20 KB</td></tr><tr><td>16 KB</td></tr><tr><td>8 KB</td></tr><tr><td>2 KB</td></tr><tr><td>6 KB</td></tr></table>	OS	4 KB	9 KB	20 KB	16 KB	8 KB	2 KB	6 KB	Worst fit: Allocate the largest free block to the new process <table><tr><td>OS</td></tr><tr><td>4 KB</td></tr><tr><td>9 KB</td></tr><tr><td>20 KB</td></tr><tr><td>16 KB</td></tr><tr><td>8 KB</td></tr><tr><td>2 KB</td></tr><tr><td>6 KB</td></tr></table>	OS	4 KB	9 KB	20 KB	16 KB	8 KB	2 KB	6 KB																																			
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5.	a) Ans.	Attempt any <u>TWO</u> of the following: Write two uses of the following operating system tools: i) Security Policy ii) User Management iii) Performance Monitor <u>i) Security Policy:</u> -As OS security policies and procedures cover a large area, there are following techniques to addressing them: • Installing and updating anti-virus software • Ensure the systems are patched or updated regularly \$ yum updates \$yum check-update • Implementing user management policies to protect user accounts and privileges. • Installing a firewall and ensuring that it is properly set to monitor all incoming and outgoing traffic. - Use Secure SSH: Telnet and rlogin protocols uses plain text, not encrypted format which is the security breaches. SSH is a secure protocol that use encryption technology during communication with server. Never login directly as root unless necessary. Use “sudo” to execute commands. \$ vi /etc/ssh/sshd_config \$ PermitRootLogin no #disable root login \$AllowUsers username # Allow only specific users \$ Protocol 2 #use SSH protocol 2 version Lock and Unlock Features: They are very useful, instead of removing an account from the system, you can lock it for an week or a month. To lock a specific user, you can use the follow command. \$ passwd -l accountname Turn Off IPv6: If you are not using a IPv6 protocol, then you should disable it because most of the applications or policies not required IPv6 protocol and currently it doesn't required on the server. Go to network configuration file and add followings lines to disable it. # vi /etc/sysconfig/network NETWORKING_IPV6=no IPV6INIT=no	12 6M <i>Each tool with any 2 valid uses 2M</i>
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	Enables Iptables (Firewall): It is highly recommended to enable Linux firewall to secure unauthorised access of your servers. Apply rules in iptables to filters incoming, outgoing and forwarding packets. We can specify the source and destination address to allow and deny in specific udp/tcp port number. Use Strong Password Policy: Passwords are always a security problem because humans are. People often cannot be bothered to come up with a lot of different passwords, so they use the same ones in different places or combinations that are easy to remember, like “password” or “abcde”. Basically, a gift to hackers. Make it a requirement that any password must contain both upper and lower case, be a mix of numbers, letters and symbols and you’ll be way safer. <u>ii) User Management:</u> -As the administrator, it is your job to create and manage the accounts for all required users. -User management includes everything from creating a user to deleting a user on your system and also assigning the passwords to created users using “passwd” command. Following are the Linux command line tools for managing users and groups: 1. useradd. 2. usermod. 3. userdel. <u>iii) Performance Monitor</u> It is really very tough job for every System or Network administrator to monitor and debug Linux System Performance problems every day. • The top command used to display all the running and active real-time processes in ordered list and updates it regularly. It display CPU usage, Memory usage, Swap Memory, Cache Size, Buffer Size, Process PID, User, Commands and much more. It also shows high memory and CPU utilization of a running processes. • Linux vmstat command used to display statistics of virtual memory, kernel threads, disks, system processes, I/O blocks, interrupts, CPU activity and much more.	
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b) Ans.	Differentiate between process and thread (any two Points). Also discuss the benefits of multithreaded programming.			6M
		Process	Thread	<i>Differentiate - any 2 valid points</i> 2M <i>Benefits</i> 4M
	Definition	A process is a program under execution i.e. an active program.	A thread is a lightweight process that can be managed independently by a scheduler.	
	Running mechanism	Processes run in separate memory spaces.	Threads within the same process run in a shared memory space.	
	Concept	Process is heavy weight and any program is in execution.	It is a lightweight process and a segment of a process.	
	PCB	The process has its own Process Control Block, Stack, and Address Space.	Thread has Parents' PCB, its own Thread Control Block, and Stack and common Address space.	
	Context Switching	Context switching between the process is more expensive	Context switching between threads of the same process is less expensive	
	Dependency	Processes are independent.	Threads are dependent	
	Controlling	Process is controlled by the operating system.	Threads are controlled by programmer in a program	
	Benefits of multithreaded programming: The benefits of multithreaded programming can be broken down into four major categories: Responsiveness: Multithreading an interactive application may allow a program to continue running even if part of it is blocked or is performing a lengthy operation, thereby increasing responsiveness to the user. For			



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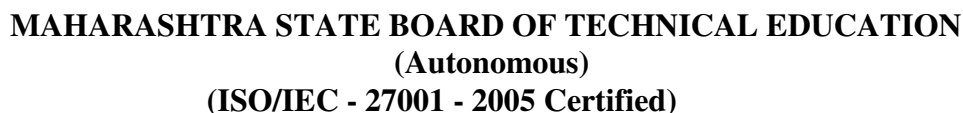
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		instance, a multithreaded Web browser could allow user interaction in one thread while an image was being loaded in another thread. Resource sharing: Processes may only share resources through techniques such as shared memory or message passing. Such techniques must be explicitly arranged by the programmer. However, threads share the memory and the resources of the process to which they belong by default. The benefit of sharing code and data is that it allows an application to have several different threads of activity within the same address space. Economy: Allocating memory and resources for process creation is costly. Because threads share the resources of the process to which they belong, it is more economical to create and context-switch threads. Empirically gauging the difference in overhead can be difficult, but in general it is much more time consuming to create and manage processes than threads. In Solaris, for example, creating a process is about thirty times slower than is creating a thread, and context switching is about five times slower. Scalability: The benefits of multithreading can be greatly increased in a multiprocessor architecture, where threads may be running in parallel on different processors. A single-threaded process can only run on one processor, regardless how many are available. Multithreading on a multi-CPU machine increases parallelism.	
	c)	Find out the total number of page faults using: i. Least Recently used Page Replacement ii. Optimal Page Replacement Page replacement algorithms of memory management, if the pages are coming in the order: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1	6M



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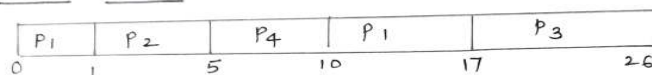
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- Preemptive scheduling allows a process to be interrupted in the midst of its execution, taking the CPU away and allocating it to another process. Non-preemptive scheduling ensures that a process relinquishes control of the CPU only when it finishes with its current CPU burst.

* Pre-emptive SJF :-

Process	Arrival Time	Burst Time	Waiting Time (ms)	Turn - Around Time (ms)
P ₁	0	8	10-1 = 9	17-0 = 17
P ₂	1	4	1-1 = 0	5-1 = 4
P ₃	2	9	17-2 = 15	26-2 = 24
P ₄	3	5	5-3 = 2	10-3 = 7

• Gantt chart :-



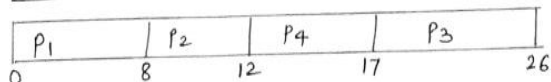
• Average Waiting Time = $\frac{9+0+15+2}{4} = \frac{26}{4} = 6.5 \text{ ms}$

• Average Turn-Around Time = $\frac{17+4+24+7}{4} = \frac{52}{4} = 13 \text{ ms}$

* Non-Pre-emptive SJF :-

Process	Arrival Time	Burst Time	Waiting Time (ms)	Turn - Around Time (ms)
P ₁	0	8	0	8-0 = 8
P ₂	1	4	8-1 = 7	12-1 = 11
P ₃	2	9	17-2 = 15	26-2 = 24
P ₄	3	5	12-3 = 9	17-3 = 14

• Gantt chart :-



• Average Waiting Time = $\frac{0+7+15+9}{4} = \frac{31}{4} = 7.75 \text{ ms}$

• Average Turn-Around Time = $\frac{8+11+24+14}{4} = \frac{57}{4} = 14.25 \text{ ms}$



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	b) Ans.	List free space management technique with the help of neat diagram, explain any one technique in detail. A file system is responsible to allocate the free blocks to the file therefore it has to keep track of all the free blocks present in the disk. There are mainly two approaches by using which, the free blocks in the disk are managed. Following are the free space management technique: 1) Bitmap 2) Linked List 3) Grouping Bit Vector The first method that we will discuss is the bit vector method. Also known as the bit map, this is the most frequently used method to implement the free space list. In this method, each block in the hard disk is represented by a bit (either 0 or 1). If a block has a bit 0 means that the block is allocated to a file, and if a block has a bit 1 means that the block is not allocated to any file, i.e., the block is free. For example, consider a disk having 16 blocks where block numbers 2, 3, 4, 5, 8, 9, 10, 11, 12, and 13 are free, and the rest of the blocks, i.e., block numbers 0, 1, 6, 7, 14 and 15 are allocated to some files. The bit vector for this disk will look like this- <table><tr><td>Blocks</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>Bits</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> Block is allocated Block is free We can find the free block number from the bit vector using the following method- Block number = (Number of bits per word)*(number of 0-value words) + (offset of first bit) We will now find the first free block number in the above example. The first group of 8 bits (00111100) constitutes a non-zero word since all bits are not 0. After finding the non-zero word, we will look for the first 1 bit. This is the third character of the non-zero word. Hence, offset = 3. Therefore, the first free block number = 8 * 0 + 3 = 3. Linked List: Another method of doing free space management in operating systems is a linked list. In this method, all the free blocks existing in the disk are linked together in a linked list. The address of the first	Blocks	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Bits	0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0	6M <i>Listing free space management</i> 1M <i>Explanation</i> 3M <i>Diagram</i> 2M
Blocks	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15																					
Bits	0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0																					



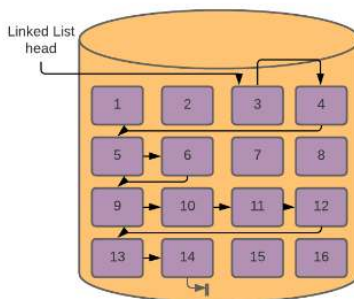
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free block is stored somewhere in the memory. Each free block contains a pointer that contains the address to the next free block. The last free block points to null, indicating the end of the linked list. For example, consider a disk having 16 blocks where block numbers 3, 4, 5, 6, 9, 10, 11, 12, 13, and 14 are free, and the rest of the blocks, i.e., block numbers 1, 2, 7, 8, 15 and 16 are allocated to some files. If we maintain a linked list, then Block 3 will contain a pointer to Block 4, and Block 4 will contain a pointer to Block 5.



Grouping: The third method of free space management in operating systems is grouping. This method is the modification of the linked list method. In this method, the first free block stores the addresses of the n free blocks. The first n-1 of these blocks is free. The last block in these n free blocks contains the addresses of the next n free blocks, and so on.

For example, consider a disk having 16 blocks where block numbers 3, 4, 5, 6, 9, 10, 11, 12, 13, and 14 are free, and the rest of the blocks, i.e., block numbers 1, 2, 7, 8, 15 and 16 are allocated to some files.

If we apply the Grouping method considering n to be 3, Block 3 will store the addresses of Block 4, Block 5, and Block 6. Similarly, Block 6 will store the addresses of Block 9, Block 10, and Block 11. Block 11 will store the addresses of Block 12, Block 13, and Block 14. This is also represented in the following figure

Block 3 -> 4, 5, 6
Block 6 -> 9, 10, 11
Block 11 -> 12, 13, 14



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	c) Ans.	Draw and explain directory structure of a file system in terms of single level, two level and tree structure. A directory is a container that is used to contain folders and files. Following are the logical structures of a directory: 1) Single level 2) Two level 3) Tree structure <u>Single-level directory:</u> The single-level directory is the simplest directory structure. In it, all files are contained in the same directory, which makes it easy to support and understand. Single level directory has a significant limitation, however, when the number of files increases or when the system has more than one user. Since all the files are in the same directory, they must have a unique name. If two users call their dataset test, then the unique name rule violated. •Simple operations like file creation, search, deletion, and updating are possible with a single-level directory structure. •The single-level directory is easier to understand in practical life. •Two file names cannot be the same. In case two files are given the same name, the previous one is overridden. •If the number of files is very large, searching a particular file is very inefficient. •Segregation of important and unimportant files is not possible. •The single-level directory is not useful for multi-user systems. Directory The diagram illustrates a single-level directory structure. At the top, there is a horizontal row of eight rectangular boxes labeled D1, D2, D3, D4, D5, D6, D7, and D8. Below each of these boxes, a vertical arrow points down to a circular box. These circular boxes are labeled f1, f2, f3, f4, f5, f6, f7, and f8. The word 'Files' is written below the circular boxes. The entire structure is labeled 'Directory' at the top and 'Files' at the bottom. Fig:Single level Directory Structure	6M <i>Relevant explanation shall be considered</i> <i>Single level Diagram 1M</i> <i>Explanation 1M</i> <i>Two Level Diagram 1M</i> <i>Explanation 1M</i> <i>Tree Level Diagram 1M</i> <i>Explanation 1M</i>
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Two-level directory:

As we have seen, a single level directory often leads to confusion of files names among different users. The solution to this problem is to create a **separate directory for each user**.

In the two-level directory structure, each user has their own **user files directory (UFD)**. The UFDs have similar structures, but each lists only the files of a single user. System's **master file directory (MFD)** is searched whenever a new user id is created.

- Searching is very easy.
- There can be two files with the same name in two different user directories. Since they are not in the same directory, the same name can be used.
- Grouping is easier.
- A user cannot enter another user's directory without permission.
- Implementation is easy.
- It does not allow users to create subdirectories.

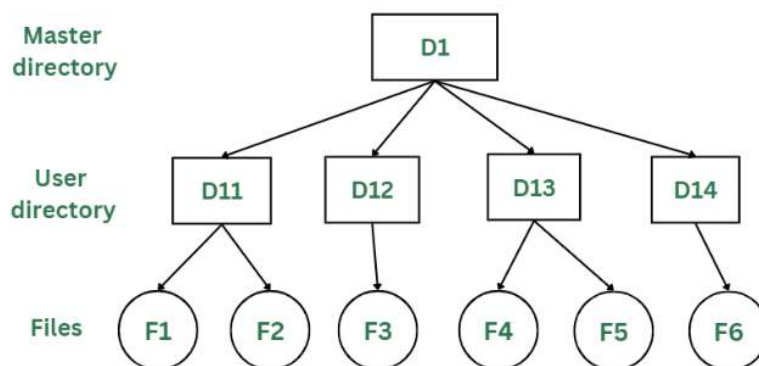


Fig: Two Level Directory structure

Tree Structure/ Hierarchical Structure:

Tree directory structure of operating system is most commonly used in our **personal computers**. User can create files and subdirectories too, which was a disadvantage in the previous directory structures.

This directory structure resembles a real tree upside down, where the **root directory** is at the peak. This root contains all the directories for each user. The users can create subdirectories and even store files in their directory.



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A user do not have access to the root directory data and cannot modify it. And, even in this directory the user do not have access to other user's directories. The structure of tree directory is given below which shows how there are files and subdirectories in each user's directory.

- The root directory is highly secured, and only the system administrator can access it. We can see how there can be subdirectories inside the user directories. A user cannot modify the root directory data. Also, a user cannot access another user's directory.
- Allows subdirectories inside a directory.
- Searching is easy.
- Allows grouping.
- Segregation of important and unimportant files is easy.
- As one user cannot enter another user's directory, this restricts sharing of files.
- Too many subdirectories may make the search complicated.
- Users cannot modify the root directory's data.

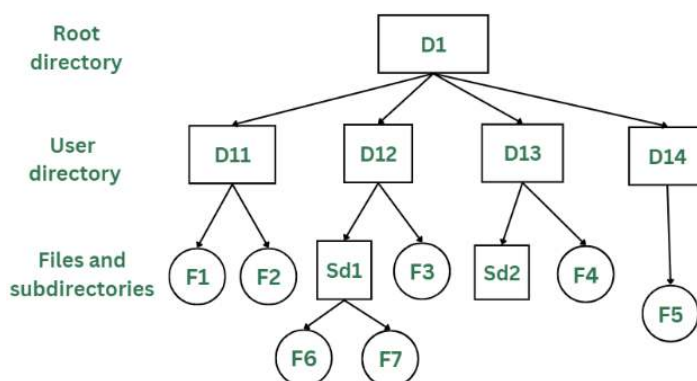


Fig: Tree Structure/ Hierarchical Structure



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Important Instructions to examiners:

- 1) The answers should be examined by key words and not as word-to-word as given in the model answer scheme.
- 2) The model answer and the answer written by candidate may vary but the examiner may try to assess the understanding level of the candidate.
- 3) The language errors such as grammatical, spelling errors should not be given more Importance (Not applicable for subject English and Communication Skills).
- 4) While assessing figures, examiner may give credit for principal components indicated in the figure. The figures drawn by candidate and model answer may vary. The examiner may give credit for any equivalent figure drawn.
- 5) Credits may be given step wise for numerical problems. In some cases, the assumed constant values may vary and there may be some difference in the candidate's answers and model answer.
- 6) In case of some questions credit may be given by judgement on part of examiner of relevant answer based on candidate's understanding.
- 7) For programming language papers, credit may be given to any other program based on equivalent concept.
- 8) As per the policy decision of Maharashtra State Government, teaching in English/Marathi and Bilingual (English + Marathi) medium is introduced at first year of AICTE diploma Programme from academic year 2021-2022. Hence if the students in first year (first and second semesters) write answers in Marathi or bilingual language (English +Marathi), the Examiner shall consider the same and assess the answer based on matching of concepts with model answer.

Q. No	Sub Q.N.	Answer	Marking Scheme
1.	a) Ans.	Attempt any <u>FIVE</u> of the following: List different types of operating system • Batch Operating System • Multi-Programming Operating System • Time Shared Operating System • Multiprocessor System • Distributed System • Real Time Operating System • Mobile Operating System	10 2M <i>Any four types 1/2M each</i>
	b) Ans.	State any four services provided by an operating system • User Interface • Program Execution • I/O Operation	2M <i>Any four services 1/2M each</i>



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		• File System Manipulation • Communication • Error Detection • Resource Allocation • Accounting • Protection and Security	
	c) Ans.	Draw process state diagram <pre> graph LR new([new]) -- admitted --> ready([ready]) ready -- scheduler dispatch --> running([running]) running -- interrupt --> ready running -- "I/O or event wait" --> waiting([waiting]) waiting -- "I/O or event completion" --> ready running -- exit --> terminated([terminated]) </pre>	2M <i>Correct diagram 2M</i>
	d) Ans.	State two features of non-preemptive scheduling • Once CPU is allocated to the process, the process keeps the CPU until it releases the CPU either by terminating or by switching to the waiting state. • A process switches from running state to waiting state or running to terminated state. • Low scheduling overhead. • Simple to implement and manage. • Less responsive in terms of priority for urgent task.	2M <i>Any two relevant features 1M each</i>
	e) Ans.	Define following terms: Memory compaction Fragmentation i. Memory compaction: The process of shuffling the memory contents so as to place all free memory together in one large block is known as memory compaction. ii. Fragmentation: When processes are loaded and removed from memory, the free memory space is broken into little pieces which is known as fragmentation.	2M <i>Correct definition 1M each</i>



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	f) Ans.	Write syntax of PWD command and explain its use with the help of suitable example. Syntax: \$ pwd PWD command is used to print (display) current working directory. Example: \$ pwd Above example displays absolute pathname. Output: /home/poly	2M <i>Syntax 1M</i> <i>Use & example 1M</i>
	g) Ans.	List any four file operations. • Creating a file • Writing a file • Reading a file • Repositioning within a file • Deleting a file • Appending new information to the end of the file • Renaming an existing file. • Truncating a file • Creating copy of a file, copy file to another I/O device such as printer or display	2M <i>Any four operations 1/2M each</i>
2.	a) Ans.	Attempt any <u>THREE</u> of the following: Explain Resource management of an operating system. • Resource Management in Operating System is the process to manage all the resources efficiently like CPU, memory, input/output devices, and other hardware resources among the various programs and processes running in the computer. • Computer's resources are limited and several users or programs may need to utilize the same resources—such as memory and CPU—at the same time. • Operating system has to manage and ensure that all processes get the resources they need to execute, without any problems like deadlocks. • Operating systems use various scheduling algorithms to ensure that each process gets its fair share of CPU time. • Operating systems use virtual memory techniques to manage memory efficiently. • Operating systems use file system management techniques to manage disk space efficiently. • File system management allows the operating system to create,	12 4M <i>Any relevant explanation 4M</i>



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		delete, and modify files and directories in Secondary Storage Device. Operating systems use various network management techniques to manage network bandwidth efficiently.	
	b) Ans.	Explain different components of operating system. List of System Components: 1. Process Management 2. Main Memory Management 3. File Management 4. I/O System Management 5. Secondary Storage Management 1.Process Management: A program is a set of instructions. When CPU is allocated to a program, it can start its execution. A program in execution is a process. A word processing program run by a user on a PC is a process. A process needs various system resources including CPU time, memory, files and I/O devices to complete the job execution. These resources can be given to the process when it is created or allocated to it while it is running. The operating system responsible for the following activities in connection with process management: • Creation and deletion of user and system processes. • Suspension and resumption of processes. • A mechanism for process synchronization. • A mechanism for process communication. • A mechanism for deadlock handling. 2. Main-Memory Management Main memory is a large array of words or bytes, ranging in size from hundreds of thousands to billions. Each word or byte has its own address. Main memory is a repository of quickly accessible data shared by the CPU and I/O devices. The central processor reads instructions from main memory during the instruction fetch cycle and both reads and writes data from main memory during the data fetch cycle. The main memory is generally the only large storage device that the CPU is able to address and access directly. The operating system responsible for the following activities in connection with main memory s management:	4M <i>Any two components with relevant explanation 2M each</i>



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	• Keeping track of which parts of memory are currently being used and by whom. • Deciding which processes (or parts thereof) and data to move into and out of memory. • Allocating and deallocating memory space as needed. 3. File Management A file is a collected of related information defined by its creator. Computer can store files on the disk (secondary storage), which provide long term storage. Some examples of storage media are magnetic tape, magnetic disk and optical disk. Each of these media has its own properties like speed, capacity, and data transfer rate and access methods. A file system normally organized into directories to ease their use. These directories may contain files and other directions. The operating system responsible for the following activities in connection with file management: • The creation and deletion of files. • The creation and deletion of directions. • The support of primitives for manipulating files and directions. • The mapping of files onto secondary storage. • The backup of files on stable storage media. 4. I/O device Management Input / Output device management provides an environment for the better interaction between system and the I/O devices (such as printers, scanners, tape drives etc.). To interact with I/O devices in an effective manner, the operating system uses some special programs known as device driver. The device drivers take the data that operating system has defined as a file and then translate them into streams of bits or a series of laser pulses (in regard with laser printer). The I/O subsystem consists of several components: • A memory management component that includes buffering, caching, spooling • A general device driver interface • Drivers for specific hardware devices	
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		5. Secondary-Storage Management The computer system provides secondary storage to back up main memory. Secondary storage is required because main memory is too small to accommodate all data and programs, and the data that it holds is lost when power is lost. Most of the programs including compilers, assemblers, word processors, editors, and formatters are stored on a disk until loaded into memory. Secondary storage consists of tapes drives, disk drives, and other media. The operating system is responsible for the following activities in connection with disk management: • Free space management • Storage allocation • Disk scheduling.	
	c) Ans.	Describe message passing system of interprocess communication (IPC) Message Passing: In this model, communication takes place by exchanging messages between cooperating processes. It allows processes to communicate and synchronize their action without sharing the same address space. It is particularly useful in a distributed environment when communication process may reside on a different computer connected by a network. Communication requires sending and receiving messages through the kernel. The processes that want to communicate with each other must have a communication link between them. Between each pair of processes exactly one communication link exists.	4M <i>Relevant explanation 3M</i> <i>Correct diagram 1M</i>



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	d) Ans.	What is CPU Scheduler? Explain the preemptive and non-preemptive type of scheduling CPU Scheduler: When the CPU becomes idle, the CPU scheduler selects one of the processes in the ready queue for execution. The CPU scheduler selects a process from the processes in memory that are ready to execute and allocates the CPU to that process. Pre-emptive scheduling: - Once the CPU is allocated to a process, the CPU is forcefully removed from that process due to occurrence of interrupt or completion of I/O request by the requesting process. In this scheduling, a process changes its status from running state to ready state or waiting state to ready state. Non-Preemptive scheduling: - Once the CPU is allocated to a process, the process keeps the CPU until it releases the CPU by terminating or by switching to the waiting state. Control of CPU is with the process to which it is allocated. In this scheduling, a process changes its status from running to waiting or running to terminate.	4M CPU scheduler 2M Relevant explanation of Preemptive 1M & non-preemptive 1M																		
3.	a) Ans.	Attempt any <u>THREE</u> of the following: Define Process. Draw a Process Control Block and explain the information in PCB Process: A process is defined as, a program under execution, which competes for the CPU time and other resources. A process is a program in execution. Process is also called as job, task and unit of work. <table><tr><th>Pointer</th><th>Process State</th></tr><tr><td colspan="2">Process Number</td></tr><tr><td colspan="2">Program Counter</td></tr><tr><td colspan="2">CPU Registers</td></tr><tr><td colspan="2">Memory Allocations</td></tr><tr><td colspan="2">Event Information</td></tr><tr><td colspan="2">List of Open Files</td></tr><tr><td colspan="2">⋮</td></tr><tr><td colspan="2">•</td></tr></table> Fig: Process Control Block	Pointer	Process State	Process Number		Program Counter		CPU Registers		Memory Allocations		Event Information		List of Open Files		⋮		•		12 4M Define process 1M Diagram of PCB with correct naming / block 1M Any 4 blocks of information in PCB 2M
Pointer	Process State																				
Process Number																					
Program Counter																					
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	Following are the blocks in PCB: 1. Process Number: Each process is identified by its process number, called Process Identification Number (PID). Every process has a unique process-id through which it is identified. The process-id is provided by the OS. The process id of two processes could not be same because process-id is always unique. 2. Priority: Each process is assigned a certain level of priority that corresponds to the relative importance of the event that it services process priority is the preference of the one process over other process for execution. Priority may be given by the user/system manager or it may be given internally by OS. This field stores the priority of a particular process. 3. Process State: This information is about the current state of the process. The state may be new, ready, running, and waiting, halted, and so on. 4. Program Counter: The counter indicates the address of the next instruction to be executed for this process. 5. CPU Registers: The registers vary in number and type, depending on the computer architecture. They include accumulators, index registers, stack pointers, and general-purpose registers, plus any condition-code information. Along with the program counter, this state information must be saved when an interrupt occurs, to allow the process to be continued correctly afterward. 6. CPU Scheduling Information: This information includes a process priority, pointers to scheduling queues, and any other scheduling parameters. 7. Memory Management Information: This information may include such information as the value of the base and limit registers, the page tables, or the segment tables, depending on the memory system used by the operating system.	
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		8. Accounting Information: This information includes the amount of CPU and real time used, time limits, account numbers, job or process numbers, and so on. 9. I/O Status Information: This information includes the list of I/O devices allocated to the process, a list of open files, and so on. 10. File Management: It includes information about all open files, access rights etc. 11. Pointer: Pointer points to another process control block. Pointer is used for maintaining the scheduling list.	
	b) Ans.	Define deadlock. State the conditions necessary for deadlock. Deadlock: Deadlock is a situation when two or more processes get locked and cannot processed further because of inter-dependability. In real world, deadlocks can arise when two persons wait for phone calls from one another. OR Deadlock is defined as, "a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process". Necessary Conditions to Deadlock: Mutual Exclusion: At least one resource is held in a non-sharable mode, that is only one process at a time can use the resource. If another process requests that resource, the requesting process must be delayed until the resource has been released. Each resource is either currently assigned to exactly one process or is available. Hold and Wait: There must exist a process that is holding at least one resource and is waiting to acquire additional resources that are currently being held by another process. Process currently holding resources granted earlier can request new resources. No Pre-emption: Resources cannot be pre-empted; i.e. resource can only be released voluntarily by the process holding it, after the process has completed its task. Resources previously granted cannot be forcibly taken away from a process. They must be explicitly released by the process holding them.	4M <i>Define deadlock 1M</i> <i>State any 3 conditions necessary for deadlock 3M</i>



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		Circular Wait: There exist a set (P0, P1, ----- Pn) of waiting processes such that P0 is waiting for a resource which is held by P1, P1 is waiting for a resource which is held by P2. Pn-1 is waiting for resources which are held by Pn and Pn is waiting for a resource which is held by P0. Thus there must be a circular chain of two or more processes, each of which is waiting for a resource held by the next member of the chain.	
	c) Ans.	Explain the following terms with respect to memory management: i) Dynamic relocation ii) Swapping i) Dynamic Relocation When a program gets swapped out to a disk memory, then it is not always possible that when it is swapped back into main memory then it occupies the previous memory location, since the location may still be occupied by another process. We may need to relocate the process to a different area of memory. Thus there is a possibility that program may be moved in main memory due to swapping. ii) Swapping Swapping is mechanism in which a process can be swapped temporarily out of main memory (or move) to secondary storage (disk) and make that memory available to other processes. At some later time, the system swaps back the process from the secondary storage to main memory. Fig: swapping in memory management	4M <i>Each term explanation 2M</i>



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	d) Ans.	With suitable diagram, explain how contiguous file allocation is performed? • The contiguous allocation method requires each file to occupy a set of contiguous addresses on the disk. Disk addresses define a linear ordering on the disk. Contiguous allocation of a file is defined by the disk address of the first block and its length. If the file is ‘n’ blocks long and starts at location ‘b’, then it occupies blocks b, b+1, b+2, - - - - b+n-1. The directory entry for each file indicates the address of the starting block and the length of the area allocated for this file. • Contiguous allocation supports both sequential and direct access. • For direct access to block ‘i’ of a file, which starts at block ‘b’, we can immediately access block b+i. The difficulty with contiguous allocation is finding space for a new file. • For direct access to block ‘i’ of a file, which starts at block ‘b’, we can immediately access block b+i. • The difficulty with contiguous allocation is finding space for a new file. • If file to be created are ‘n’ blocks long, we must search free space list for ‘n’ free contiguous blocks. count 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 Directory <table><tr><th>File</th><th>Start</th><th>Length</th></tr><tr><td>count</td><td>0</td><td>2</td></tr><tr><td>tr</td><td>14</td><td>3</td></tr><tr><td>mail</td><td>19</td><td>6</td></tr><tr><td>list</td><td>28</td><td>4</td></tr><tr><td>f</td><td>6</td><td>2</td></tr></table>	File	Start	Length	count	0	2	tr	14	3	mail	19	6	list	28	4	f	6	2	4M Any relevant Diagram 1M Explanation 3M
File	Start	Length																			
count	0	2																			
tr	14	3																			
mail	19	6																			
list	28	4																			
f	6	2																			
4.	a)	Attempt any THREE of the following: Compare between Time sharing operating system and multiprogramming operative system.	12 4M																		



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	Ans. <table><tr><th>Time sharing operating system</th><th>Multiprogramming operating system</th></tr><tr><td>In this time sharing Operating system many users/processes are allocated with computer resources in respective time slots.</td><td>Multiprogramming operating system allows to execute multiple processes by monitoring their process states and switching in between processes.</td></tr><tr><td>Processors time is shared with multiple users that's why it is called as time sharing operating system.</td><td>Processor and memory underutilization problem is resolved and multiple programs runs on CPU that's why it is called multiprogramming.</td></tr><tr><td>In this process, two or more users can use a processor in their terminal.</td><td>In this, the process can be executed by a single processor.</td></tr><tr><td>Time sharing OS has fixed time slice.</td><td>Multi-programming OS has no fixed time slice.</td></tr><tr><td>Time sharing system minimizes response time.</td><td>Multiprogramming system maximizes processor use.</td></tr><tr><td>Example: Windows NT.</td><td>Example: Mac OS.</td></tr></table>	Time sharing operating system	Multiprogramming operating system	In this time sharing Operating system many users/processes are allocated with computer resources in respective time slots.	Multiprogramming operating system allows to execute multiple processes by monitoring their process states and switching in between processes.	Processors time is shared with multiple users that's why it is called as time sharing operating system.	Processor and memory underutilization problem is resolved and multiple programs runs on CPU that's why it is called multiprogramming.	In this process, two or more users can use a processor in their terminal.	In this, the process can be executed by a single processor.	Time sharing OS has fixed time slice.	Multi-programming OS has no fixed time slice.	Time sharing system minimizes response time.	Multiprogramming system maximizes processor use.	Example: Windows NT.	Example: Mac OS.	Any 4 correct points 1M each
Time sharing operating system	Multiprogramming operating system															
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Time sharing system minimizes response time.	Multiprogramming system maximizes processor use.															
Example: Windows NT.	Example: Mac OS.															
b) Ans.	Explain any four types of system call. 1.Process and Job Control • A running program needs to be able to halt its execution either normally (end) or abnormally (abort). If the program discovers an error in its input and wants to terminate abnormally, it may also want to define an error level. • A process or job executing one program may want to load and execute another program. This allows the control card interpreter to execute program as directly by the control cards of the user job. If we create a new job or process, we should able to control its execution. • We may also want to terminate a job or process that we created (terminate process). If we find that it is incorrect or no longer needed we need waiting time to finish execution (wait time). Another set of system calls are helpful in debugging a program.	4M Explanation of any 4 system calls 1M each														



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	Example of Process related system call: - End, Abort. - Load, Execute. - Create process, Terminate process. - Ready process, Dispatch process. - Suspend process, Resume process. - Get process attributes, Set process attributes. - Wait for Time. - Wait event, Signal event. - Change priority of process. 2.File Management - We can identify several common system calls dealing with files. We first need to be able to create and delete files such a system call requires the name of the file and perhaps some of its attributes. - Once, the file is created, we need to open it and use it. We may also read, write and reposition. Finally we need to close the file, indicating that we are no longer using it. Example of file related system call: - Create file, Delete file - Open a file, Close a file. - Create directory. - Read, Write, Reposition. - Get file attributes, Set file attributes. - Create a link. - Change working directory. 3.Device Management - Files can be thought of as abstract or virtual devices. Thus many of the system calls for files are also needed for devices. - If there are multiple users of the system however we must first request the device to ensure that we have exclusive use of it. After we are finished with the device, we must release it. These functions are similar to the open/close system calls for files. - Once, the device has been requested we can read, write and reposition the device just as with files.	
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		System call related to device management: - Request device, Release device. - Read, Write, Reposition. - Get device attributes, set device attributes. 4. Information Maintenance - Many system calls exist simply for the purpose of transferring information between the user program and the operating system. For example most systems have a system call to return the current time and date. - Other system calls may return information about the system such as the number of current users, the version number of the operating system, the amount of free memory or disk space and so on. - In addition the operating system keeps information about all of its jobs and processes and there are system calls to access this information. Generally, there are also calls to reset it. (get process attributes and set process attributes). Information Maintenance related system call: - Get Time or Date, Set Time or Date. - Get system Data, Set system Data. - Get process, File or Device attributes. - Set process, File or Device attributes. 5. Communication - There are two models of inter-process communication, the message-passing model and the shared memory model. - Message-passing uses a common mailbox to pass messages between processes - Shared memory use certain system calls to create and gain access to regions of memory owned by other processes Communication related system calls: - Establish a connection. - Send, receive messages. - Terminate a connection	
	c) Ans.	Describe how context switch is executed by operating system When CPU switches to another process, the system must save the state of the old process and load the saved state for the new process. This task is known as a context switch.	4M <i>Diagram</i> 1M



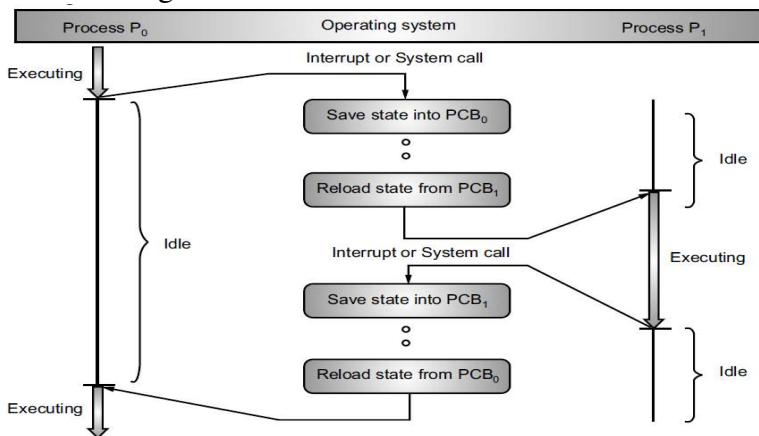
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	• CPU switching from one process to another process is called a context switch. • Context switch times are highly dependent on hardware support. Its speed varies from machine to machine, depending on the memory speed, the number of registers that must be copied and the existence of special instructions. • This enables multiple processes to share a single CPU. The context switch is an essential feature of a multitasking operating system. • When the process is switched, the following information is stored: • Program Counter • Scheduling Information • Base and limit register value • Currently used register • Changed State • I/O State • Accounting	<i>Explanation</i> 3M						
	Process P₀ Operating system Process P₁  Fig: Context Switching							
d) Ans.	Compare Short Job First (SJF) and Shortest Remaining Time (SRTN) scheduling algorithm (any four points) <table><tr><th>Short Job First (SJF)</th><th>Shortest Remaining Time (SRTN)</th></tr><tr><td>It is a non-preemptive algorithm.</td><td>It is a preemptive algorithm.</td></tr><tr><td>It involves less overhead than SRJF.</td><td>It involves more overheads than SJF.</td></tr></table>	Short Job First (SJF)	Shortest Remaining Time (SRTN)	It is a non-preemptive algorithm.	It is a preemptive algorithm.	It involves less overhead than SRJF.	It involves more overheads than SJF.	4M Any 4 correct points 1M each
Short Job First (SJF)	Shortest Remaining Time (SRTN)							
It is a non-preemptive algorithm.	It is a preemptive algorithm.							
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		<table><tr><td>It leads to comparatively lower throughput.</td><td>It leads to increased throughput as execution time is less.</td></tr><tr><td>It minimizes the average waiting time for each process.</td><td>It may or may not minimize the average waiting time for each process.</td></tr><tr><td>It involves lesser number of context switching.</td><td>It involves higher number of context switching.</td></tr><tr><td>Short processes are executed first and then followed by longer processes.</td><td>Shorter processes run fast and longer processes show poor response time.</td></tr></table>	It leads to comparatively lower throughput.	It leads to increased throughput as execution time is less.	It minimizes the average waiting time for each process.	It may or may not minimize the average waiting time for each process.	It involves lesser number of context switching.	It involves higher number of context switching.	Short processes are executed first and then followed by longer processes.	Shorter processes run fast and longer processes show poor response time.													
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Short processes are executed first and then followed by longer processes.	Shorter processes run fast and longer processes show poor response time.																						
e) Ans.	Describe variable partitioning with the help of suitable example • In variable memory partitioning the partitions can vary in number and size. In variable memory partitioning the amount of memory allocated is exactly the amount of memory a process requires. • The operating system keeps a table indicating which parts of memory are available and which are occupied. Initially all memory is available for user programs and is considered as one large block of available memory, a hole. • When a job arrives and needs memory, we search for a hole large enough for this job. If we find one, we allocate only as much as is needed, keeping the rest available to satisfy future requests. For example, assume 256K memory available and a resident monitor of 40K. This situation leaves 216K for user programs. 0 40K 256K Monitor 216K <table><tr><th colspan="3">Job Queue</th></tr><tr><th>Job</th><th>Memory</th><th>Time</th></tr><tr><td>1</td><td>60 K</td><td>10</td></tr><tr><td>2</td><td>100 K</td><td>5</td></tr><tr><td>3</td><td>30 K</td><td>20</td></tr><tr><td>4</td><td>70 K</td><td>8</td></tr><tr><td>5</td><td>50 K</td><td>15</td></tr></table>	Job Queue			Job	Memory	Time	1	60 K	10	2	100 K	5	3	30 K	20	4	70 K	8	5	50 K	15	4M Explanation 2M Any relevant Example 2M
Job Queue																							
Job	Memory	Time																					
1	60 K	10																					
2	100 K	5																					
3	30 K	20																					
4	70 K	8																					
5	50 K	15																					



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5.	a) Ans.	Attempt any <u>TWO</u> of the following: Explain the use of following OS tools: i) Device Manager ii) Task Scheduler i) Device Manager use: 1. Managing all the hardware or virtual devices of computer system. 2. Allow interaction with hardware devices through device driver. 3. Used to install device and component-level drivers as well as associated software. 4. Allocate devices to the process as per process requirement and priority. 5. Deallocate devices either temporarily or permanently depending on condition. 6. Keeping track of all device's data and location. 7. Monitoring device status like printers, storage drivers and other devices. 8. Used to enforce the predetermined policies and decides which process receives the device when and for how long. ii) Task scheduler use: 1. Used to automate the execution of programs, scripts, and various tasks at specific intervals or specific events. 2. Used to select the next jobs to be admitted into the system and the next process to run. 3. Schedules tasks on CPU. Which task to run first, which has high priority, which has low latency is decided by schedulers.	12 6M <i>Any 2 uses of Device Manager 1½ M each</i> <i>Any 2 uses of Task Scheduler 1½ M each</i>



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		4. Schedulers aim to provide fair access to system resources among competing processes or users. 5. Task schedulers strive to maximize the overall throughput of the system by executing tasks in a manner that minimizes idle time and maximizes the number of tasks completed within a given timeframe.	
	b) Ans.	Explain user level thread and Kernel level thread with its advantages and disadvantages User Level Thread • In a user thread, all of the work of thread management is done by the application and the kernel is not aware of the existence of threads. • The thread library contains code for creating and destroying threads, for passing message and data between threads, for scheduling thread execution and for saving and restoring thread contexts. • The application begins with a single thread and begins running in that thread. • User level threads are generally fast to create and manage. Advantages of user level thread over Kernel level thread: 1. Thread switching does not require Kernel mode privileges. 2. User level thread can run on any operating system. 3. Scheduling can be application specific. 4. User level threads are fast to create and manage. Disadvantages of user level thread: 1. Limited Parallelism: Since user-level threads are managed within the context of a single process, they are not well-suited for applications that need to take full advantage of multi-core processors or for applications that require parallel execution across different processes. 2. Blocking: If a user-level thread blocks due to I/O or other reasons, all threads in the same process can be blocked, leading to poor utilization of resources. 3. No True Parallelism: User-level threads cannot execute in true parallel on multi-core systems since they are managed by a single kernel-level thread	6M <i>For each Thread explanation 1M</i> <i>1 advantage of each 1M</i> <i>1 Dis-advantage of each 1M</i> <i>(Diagram optional)</i>



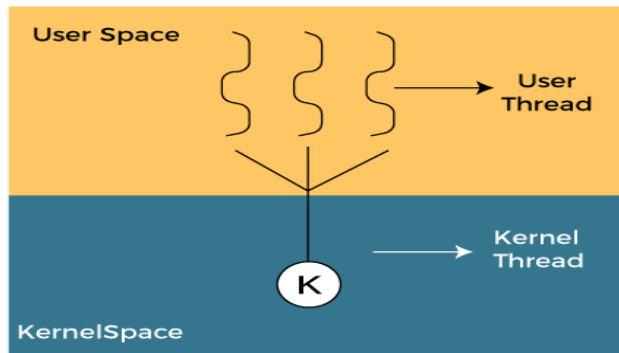
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4. It is not appropriate for a multiprocessor system.



Kernel Level Threads

- In Kernel level thread, thread management done by the Kernel.
- There is no thread management code in the application area.
- Kernel threads are supported directly by the operating system.
- Any application can be programmed to be multithreaded.
- All of the threads within an application are supported within a single process.
- The Kernel maintains context information for the process as a whole and for individual threads within the process.
- Scheduling by the Kernel is done on a thread basis.
- The Kernel performs thread creation, scheduling and management in Kernel space.
- Kernel threads are generally slower to create and manage than the user threads.

Advantages of Kernel level thread:

1. Kernel can simultaneously schedule multiple threads from the same process on multiple process.
2. If one thread in a process is blocked, the Kernel can schedule another thread of the same process.
3. Kernel routines themselves can multithreaded.

Disadvantages:

1. Kernel threads are generally slower to create and manage than the user threads.
2. Transfer of control from one thread to another within same process requires a mode switch to the Kernel.



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	c) Ans.	Consider the string: 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7 with frame size 3 and 4, calculate page fault in both the cases using FIFO algorithm <i>Note: Representation of frame can be in any order</i> FIFO : i)Frame Size=3 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6.	a)	Attempt any <u>TWO</u> of the following: What is the average turnaround time for the following process using : i) FCFS scheduling algorithm ii) SJF non preemptive scheduling algorithm iii) Round Robin Scheduling algorithm	12 6M																																																																																																																																																																																																																																																																																																																																																																																						



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Ans.	<table><tr><th>Process</th><th>Arrival time</th><th>Burst time</th></tr><tr><td>P₁</td><td>0</td><td>8</td></tr><tr><td>P₂</td><td>1</td><td>4</td></tr><tr><td>P₃</td><td>2</td><td>1</td></tr></table>	Process	Arrival time	Burst time	P ₁	0	8	P ₂	1	4	P ₃	2	1	For each algorithm: Gantt Chart 1M Correct turnaround time calculation 1M each
	Process	Arrival time	Burst time											
P ₁	0	8												
P ₂	1	4												
P ₃	2	1												
Note :Any time slice can be considered by student for Round Robin e.g. 2, 3, 4 etc i) <u>Gantt Chart FCFS</u> <table><tr><td>P₁</td><td>P₂</td><td>P₃</td></tr><tr><td>0</td><td>8</td><td>12 13</td></tr></table> Turnaround Time =Completion Time - Arrival Time P₁=8-0=8 P₂=12-1=11 P₃=13-2=11 Average turnaround time=Turnaround time of all processes / Number of processes = (8+11+11) /3 =10 ms / Units ii) <u>Gantt Chart SJF non-preemptive</u> <table><tr><td>P₁</td><td>P₃</td><td>P₂</td></tr><tr><td>0</td><td>8 9</td><td>13</td></tr></table> Turnaround Time =Completion Time - Arrival Time P₁=8-0=8 P₂=13-1=12 P₃=9-2=7 Average turnaround time=Turnaround time of all processes / Number of processes = (8+12+7) /3 = 9 ms/ Units	P ₁	P ₂	P ₃	0	8	12 13	P ₁	P ₃	P ₂	0	8 9	13		
P ₁	P ₂	P ₃												
0	8	12 13												
P ₁	P ₃	P ₂												
0	8 9	13												



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	iii) <u>Gantt Chart Round Robin</u> <u>Consider Time Quantum= 2 Units</u> <table><tr><td>P₁</td><td>P₂</td><td>P₃</td><td>P₁</td><td>P₂</td><td>P₁</td><td>P₁</td></tr><tr><td>0</td><td>2</td><td>4</td><td>5</td><td>7</td><td>9</td><td>11</td><td>13</td></tr></table> Turnaround Time =Completion Time - Arrival Time P₁=13-0=13 P₂=9-1=8 P₃=5-2=3 Average turnaround time=Turnaround time of all processes / Number of processes 	P ₁	P ₂	P ₃	P ₁	P ₂	P ₁	P ₁	0	2	4	5	7	9	11	13
P ₁	P ₂	P ₃	P ₁	P ₂	P ₁	P ₁										
0	2	4	5	7	9	11	13									



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For example, consider a disk where blocks 2, 3, 4, 5, 8, 9, 10, 11, 12, 13 are free and the rest of the blocks are allocated.

The free-space bit map would be : 001111001111100

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	1	1	1	1	0	0	1	1	1	1	1	1	0	0

1=Free block 0= Allocated block

The main advantage of this approach is its relative simplicity and its efficiency in finding the first free block or n consecutive free blocks on the disk.

Advantages of Bit Map vector method

1. Simple and easy to understand.
2. Consumes less memory.
3. It is efficient to find free space.

Disadvantages of the Bit Map vector method

1. The operating system goes through all the blocks until it finds a free block.
2. It is not efficient when the disk size is large.

2) Linked List

In this approach, the free disk blocks are linked together i.e. a free block contains a pointer to the next free block. The block number of the very first disk block is stored at a separate location on disk and is also cache in memory. In this approach, link all the disk blocks together, keeping a pointer to the first free block. This block contains a pointer to the next free disk block, and so on.

*Any 1
Advantage
1M,*

*Any 1
Disadvantage
1M*

*Linked List
explanation
1M*

*Any 1
Advantage
1M*

*Any 1
Disadvantage
1M*

*(diagram is
optional)*



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		Advantages of linked list: 1. Files can be extended, no fragmentation problems. 2. Sequential Access is easy: just chase links. 3. No extra space for the free list. 4. Starting and ending of the file can be easily found out by the linked list. Disadvantages of linked list: 1. Cannot be effectively used for direct access method. 2. Random access is virtually impossible. Lots of seeking, even in sequential access. 3. Pointers which are required need additional memory space. 4. It is not reliable as the blocks are scattered on the disk anywhere.	
	c) Ans.	Explain with diagram single level directory structure and two level directory structure with its advantages and disadvantages. Single Level Directory Structure: This is the simplest directory structure. All the files are stored in the same directory which is easy to support and understand. But a unique name must be assigned to each file. If the number of files in the directory increases, it may become difficult to remember names of all files. Advantage: • Less time required to search the file, because all files are stored in same directory. • Single level directory structure is easy to implement and maintain. • Single level directory structure, the operations like creation, searching, deletion, updating are very easy and faster	6M <i>Single level directory structure explanation 1M,</i> <i>Any 1 Advantage 1M</i>



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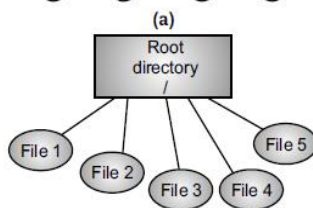
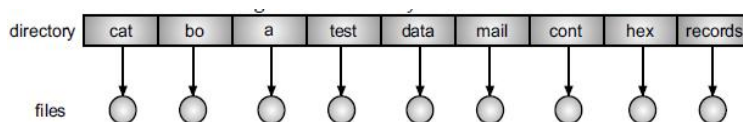
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Disadvantages

- Not suitable for a large number of files and more than one user.
- Because of single directory files, files require unique file names.
- Difficult to remember names of all the files as the number of files increases. MS-DOS OS allows only 11 character file name where as UNIX allows 255 character.



Single Level Directory Structure

Two-level directory structure: -In the two-level structures, each user has its own user file directory (UFD). The UFD lists only files of a single user. System contains a master file directory (MFD) which is indexed by user name or account number. Each entry in MFD points to the UFD for that user.

When a user refers to a particular file, only his own UFD is searched. Different users can have files with the same name, as long as all the file names within each UFD are unique.

When we create a file for a user, operating system searches only that user's UFD to find whether same name file already present in the directory. For deleting a file again operating system checks the file name in the user' UFD only.

*Any 1
disadvantage 1M*

*Two level
directory
structure
explanation
1M*

*Any 1
Advantage
1M*

*Any 1
disadvantage 1M*

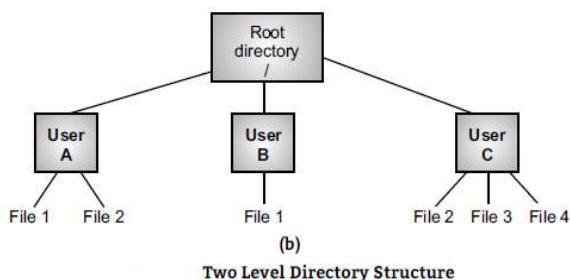


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Advantages:

- It solves isolation. i.e. data of one user is separated from other user.
- Searching is also efficient in this type of structure.
- Every user accesses his own directory and files.
- There is no problem of files with same names in different directories.

Disadvantages:

- Two users can not share data or communicate with each other.
- It is difficult to search the files with same names in different directories.

Scheme - I

Sample Question Paper

Program Name : Diploma in Engineering Group

Program Code : CO / CM / CW / IF

Semester : Fifth

Course Title : Operating System

Marks : 70

22516

Time: 3 Hrs.

Instructions:

- (1) All questions are compulsory.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data if necessary.
- (5) Preferably, write the answers in sequential order.

Q.1) Attempt any FIVE of the following.

10 Marks

- a) Differentiate between Batch Operating System and Time shared Operating System.
(any two points)
- b) State any four services of Operating System.
- c) Define : Process , Program
- d) State two features of preemptive scheduling.
- e) Define following terms
 - i) Page fault
 - ii) Segmentation
- f) Write syntax of ps command and explain its use with the help of suitable example.
- g) List any four file attributes.

Q.2) Attempt any THREE of the following.

12 Marks

- a) Explain dual modes of operations of an Operating system.
- b) Describe essential activities done by an Operating System for protection and sharing.
- c) State what is interprocess communication and explain its advantages..
- d) Describe different scheduling criteria.

Q.3) Attempt any THREE of the following.

12 Marks

- a) Define PCB. List information contained in PCB and explain any two .
- b) Define deadlock and state the necessary conditions for deadlock..
- c) Explain following terms with respect to Memory management :
 - I. Compaction
 - II. Swapping.
- d) Enlist different file allocation methods and explain any two.

Q.4) Attempt any THREE of the following.

12 Marks

- a) Compare between Windows and LINUX Operating System.(any four points)
- b) Write any four system calls related to device management.

- c) Compare between short term and long term scheduler.(any four points)
- d) Compare FCFS and SJF Scheduling algorithm with any four points.
- e) Describe contiguous memory allocation done by Operating System with the help of suitable example.

Q.5) Attempt any TWO of the following.

12 Marks

- a) Write two uses of following Operating System tools :
 - i. Performance Monitor
 - ii. Task scheduler
 - iii. User Management
- b) Write the output of the following commands.
 - a. Kill 9042018
 - b. Ps 07121975
 - c. Sleep 05
- c) Given a page reference string(arrival) with four page frames, calculate the page faults with FIFO and LRU page replacement algorithms respectively :
12, 3, 4, 5, 1, 2, 5, 1, 2, 3, 4, 5, 1,6,7,8,7,8,9,7,8,9,5,4,4,5,4,2.

Q.6) Attempt any TWO of the following.

12 Marks

- a) Solve given problem by Using FCFS to calculate average waiting time and turnaround time.

Process	Arrival time	Burst time
P1	0	7
P2	1	4
P3	2	9
P4	3	6
P5	4	8

- b) Compare between bitmap and linked list free space management techniques.(any six points)
- c) Construct and explain directory structure of a file system in terms of single level, two level and tree structure.

Scheme - I

Sample Test Paper - I

Program Name : Diploma in Engineering Group

Program Code : CO / CM / CW / IF

Semester : Fifth

Course Title : Operating System

Marks : 20

22516

Time: 1 Hour

Instructions:

- (1) All questions are compulsory.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data if necessary.
- (5) Preferably, write the answers in sequential order.

Q1. Attempt Any FOUR

08 Marks

- a) Differentiate between multiprogramming and multiprocessing. (Any 2 Points)
- b) Draw Layered structure of operating system.
- c) List any four types of OS.
- d) State any two activities performed by file management component of an Operating Systems
- e) List any four operating system services.
- f) Draw process state transition diagram .

Q2. Attempt any THREE

12 Marks

- a) Explain real time system. List its any four application.
- b) What is the purpose of the system calls? State any two calls with its functions.
- c) Describe process control block with the help of suitable diagram.
- d) Differentiate between Long term scheduler and Short term scheduler w.r.t. following points:
 - i) Selection of job
 - ii) Frequency of execution
 - iii) Speed
 - iv) Accessing which part of system

Scheme - I

Sample Test Paper - II

Program Name : Diploma in Engineering Group

Program Code : CO / CM / CW / IF

Semester : Fifth

Course Title : Operating System

Marks : 20

22516

Time: 1 Hour

Instructions:

- (1) All questions are compulsory.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data if necessary.
- (5) Preferably, write the answers in sequential order.

Q1. Attempt Any FOUR

08 Marks

- a) Describe CPU burst cycle and I/O burst cycle.
- b) What is deadlock? Give characteristics of deadlock.
- c) Describe Multilevel queue scheduling with labelled diagram.
- d) Define the term swapping.
- e) Explain different process scheduling criteria.
- f) List the directory structures.

Q2. Attempt any THREE

12 Marks

- a) With suitable example describe how to use bit map method for free space management.
- b) State the syntax and use of following process related commands :
 - i. bg
 - ii kill
- c) Describe working of contiguous file allocation method.
- d) Calculate average waiting time with Round Robin for following processes in memory. (Time Slice = 4ms)

Process	Burst time
P1	3
P2	5
P3	7
P4	4